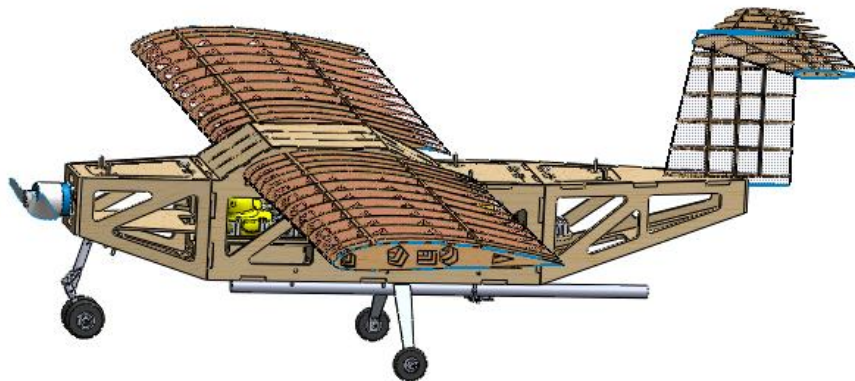


THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MECHANICAL ENGINEERING



MECH 4427 Innovative Design and Unmanned Aerial Systems
Design Report



Team Virtus

AIAA 2026 – Group 1	
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Abbreviations

AIAA	American Institute of Aeronautics and Astronautics	HKU	The University of Hong Kong
CFD	Computational Fluid Dynamics	CAD	Computer Aided Design
FEA	Finite Element Analysis	MTOW	Maximum-Take-Off Weight
C_L	Coefficient of Lift	C_D	Coefficient of Drag
C_m	Coefficient of Moment	AoA	Angle of Attack
c.g.	Centre of Gravity	DBF	Design, Build and Fly

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1. Executive Summary (Prepared by Lo Tsz Kin)

This design reports details the aerodynamic and structural design, manufacturing and testing of a banner towing bush plane from Group 1, The University of Hong Kong. We are pleased to offer Team Virtus as the best solution to this American Institute of Aeronautics and Astronautics (AIAA)'s Undergraduate Challenge. Team Virtus was designed to perform all three Missions from the competition. The first flight mission (M1) is the test flight that require the aircraft to complete 3 laps with no payload. The second flight mission (M2) is the charter flight that require the aircraft perform flying mission with as much completed lap as possible carrying predetermined number of passengers and/or cargo. The third mission (M3) is the banner flight that the aircraft is required to perform a banner deployment and release action during the flight by remote command. All three missions above must be finished within 5 minutes. The ground mission (GM) is the timed aircraft assembly and demonstration of the aircraft.

2. Introduction (Prepared by Lo Tsz Kin)

Banner towing business is a unique and memorable marking strategy for companies, standing out from traditional methods. The plane with a large banner is clearly visible and attractive to people. This form of marketing is especially useful in high traffic-areas. Thus, we created Team Virtus that can release and fly a large banner, as well as carry a minimum of three passengers. This intensive project builds real-world aerospace skills (aerodynamics, performance prediction, CAD, analysis) by completing half an aircraft lifecycle — design through manufacturing — in just two months, while delivering multiple high-performance alternative designs to the HKU DBF Team.

3. Management Summary (Prepared by Tsang Siu Tung)

3.1 Team Organization and Roles

This group has divided the design among the 6 members, as shown in Figure 1. The group leader sets weekly goals for each group member and holds weekly meetings to manage the overall progress. He is also responsible for presenting to our supervisor to report our interim progress.

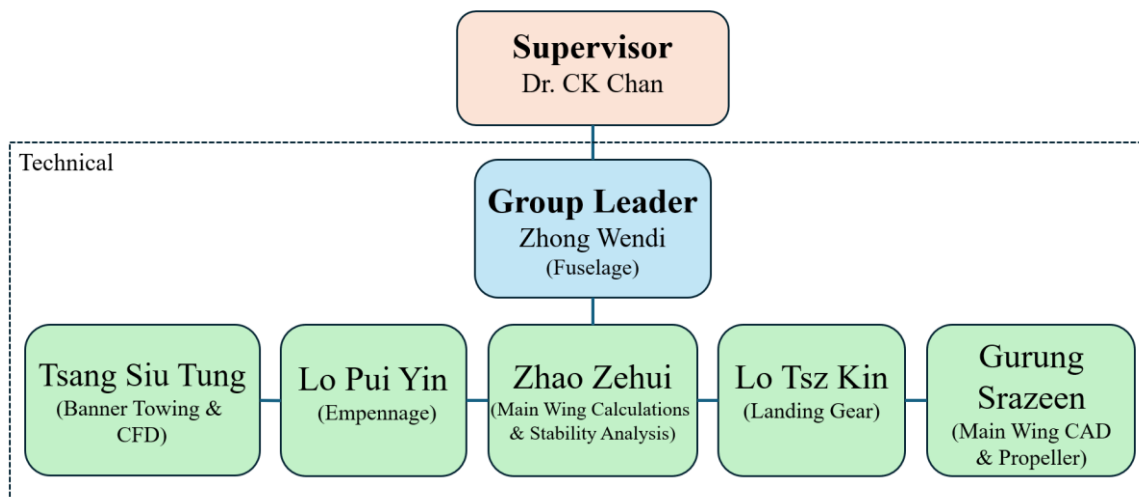


Figure 1 Group 1 Organizational Structure

Members communicate frequently to brainstorm improvements. Any big obstacles are discussed in meetings with all group members. Any stagnant period for design will cause significant delay, which may reduce the time for testing and refining on existing mechanisms.

3.2 Design Schedule

A Gantt chart was created at the beginning of October to facilitate planning and progress checking. This timeline allows individual mechanisms and 2 prototypes to be made before December. This Gantt chart is shown in Figure 2.

MECH4427 Group 1 Schedule	September				October				November			
	8	15	22	29	6	13	20	27	3	10	17	24
Review Rulebook												
Mission Sensitivity Studies												
Aerodynamics												
CAD												
CFD												
Optimization												
Manufacturing												
Presentation												
Design Report												

Table 1 Gantt chart outlining our project schedule

3.3 Budget (Prepared by Zhao Zehui)

Category	Item	Cost Total
Manufacturing	Tools	2000 (DBF & Robocon)
Propulsion, Control, & Electronics	Servo Motor, Receiver, Transmitters, Batteries	Provided by Dr Chan
	Landing Gear	Robocon
	Misc. Electronic Components	DBF & Robocon
Structures	Prototyping Materials (Wooden Board/Carbon Fiber etc)	800
	Glues & Tapes	200
	Part Commissions	100
Mission systems	Payload & Cargos	200
	Banner materials & system	200
Total Cost		4000

Table 2 Budget

4. Conceptual Design Approach (Prepared by Tsang Siu Tung & Zhao Zehui)

4.1 Mission Requirements

For Mission 1, the plane carries no payload. For Mission 2, the plane will carry passengers and/or cargo, and a deployable banner will be released in Mission 3.

$$\text{Total Mission Score} = M1 + M2 + M3 + GM \dots\dots (1)$$

The total report score, defined by Equation 2, is 15% proposal score plus 85% design report score.

$$\text{Total Report Score} = 0.15 \times \text{Proposal Score} + 0.85 \times \text{Design Report Score} \dots\dots (2)$$

The total competition score is defined by Equation 3 as shown below. Participation score will be given on a scale of 1 to 3. Thus, the report quality is exceptionally important in contributing to a higher competition score.

$$\text{Competition Score} = \text{Total Report Score} \times \text{Total Mission Score} + \text{Participation Score}$$

4.2 Misson 1 — Test Flight

Mission 1 is a test run where no payload is required. The plane must complete 3 laps within 5 minutes. One point is awarded for a successful mission.

4.3 Misson 2 — Charter Flight (Prepared by Zhao Zehui)

The payload for this mission is passengers (2.3-inch rubber ducks) and/or cargos. This mission is to simulate passenger and cargo carrying capabilities. One point is awarded for completion and the score is as follows.

$$M2 = 1 + \frac{N_{\text{Net income}}}{\text{Maximum net income}}$$

The mass specifications of the passengers (plastic ducks) and cargo (hockey pucks) are listed below:

General Physical Characteristics of Cargos			
Diameter (in)	3	Diameter (in)	7.62
Thickness (in)	1	Thickness (in)	2.54
Weight (oz)	6	Weight (g)	170.10
General Physical Characteristics of Passengers			
Width (in)	2.0 – 2.3	Width (cm)	5.08 – 5.85
Length (in)	2.0 – 2.5	Length (cm)	5.08 – 6.35
Height (in)	Up to 2.5	Height (cm)	Up to 6.35
Weight (oz)	0.6 – 0.7	Weight (g)	17.00 – 19.85

Table 3 General Physical Characteristics

The score calculation is based as the following formulas:

$$\text{Net income} = \text{Income} - \text{Cost}$$

$$\text{Income} = (\#\text{passengers} * (\text{Ip1} + (\text{Ip2} * \#\text{laps}))) + (\#\text{cargo} * (\text{Ic1} + (\text{Ic2} * \#\text{laps})))$$

Fixed income per passenger	Ip1	6
Income per passenger per lap	Ip2	2
Fixed income per cargo	Ic1	10
Income per cargo per lap	Ic2	8

Table 4 Net Income, Income and Cost

$$\text{Cost} = \#\text{laps} * (\text{Ce} + (\#\text{passengers} * \text{Cp}) + (\#\text{cargo} * \text{Cc})) * \text{EF}$$

$$\text{EF} = \text{Efficiency Factor} = \text{Total Propulsion}$$

$$\text{Battery Capacity} = (\text{W-hrs}) / 100$$

Base operating cost per lap	Ce	10
Passenger operating cost per lap	Cp	0.5
Cargo operating cost per lap	Cc	2

Table 5 Cost Parameters per Lap

4.4 Misson 3 — Banner Flight

The payload of this mission is the banner, together with the banner release mechanism. This mission is to demonstrate banner towing ability of the plane. The banner will be released by remote command. Two points are awarded for completion, and the scoring is as follows.

$$M3 = 2 + \frac{N_{\text{number of laps} \times \text{banner length} / \text{RAC}}}{\text{Max}_{\text{number of laps} \times \text{banner length} / \text{RAC}}}$$

RAC = Rated Airplane Cost. WS = Wingspan is measured and converted into one of the marking criteria.

$$RAC = 0.05 \cdot (WS/12) + 0.75$$

4.5 Misson 3 — Banner Flight

Teams can have one more attempt for a ground demonstration of Missions 2 and 3. The score is based on the mission time.

$$GM = \frac{\text{Min}_{\text{mission time}}}{N_{\text{mission time}}}$$

4.6 Conceptual Design Approach on Mission Requirements

The below table analysed our strategy to achieve a higher total mission score by compromising the maximum scores we may attain for individual missions.

Mission	Scoring	Description	Subsystem Requirements
M1	$M1 = 1.0$	- Ready aircraft within 5 minutes. - Complete three laps within 5 minutes	The aircraft must be durable and reliable to achieve stable flight under varying conditions.
M2	$1 + \frac{(Net_{income})_{Team}}{(Net_{income})_{Max}}$	- Attach payload and ready aircraft within 5 minutes. - Carry passengers and cargo onboard.	- Aircraft must have easy-to-access cabins and minimise lap time. - Optimization of the number of cargos and passengers to attain a high score
M3	$2 + \frac{\left[\frac{Laps\ Flown \times B_L}{RAC} \right]_{Team}}{\left[\frac{Laps\ Flown \times B_L}{RAC} \right]_{Max}}$	- Assemble and checkout in 5 mins. - Release banner by remote command after crossing the finish line on the last lap. - Fly as fast as possible	- Banner mechanism must work properly to stow, deploy, and release the banner. - Aircraft must take account of the drag induced by banner.
GM	$\frac{(Mission\ time)_{min}}{(Mission\ time)_{Team}}$	- Ready control surfaces - Load payload as fast as possible - Perform banner towing mechanism successfully	Beware of the mechanism control and time for the payload loading and unloading process.
Total score: $(M1 + M2 + M3 + GM) \times (Report \times 85\% + Proposal \times 15\%) + Participation\ Score$			

Table 6 Mission scoring analysis

5. Sensitivity analysis (Prepared by Zhong Wendi & Zhao Zehui)

5.1 Section overview

This section analysis how the theoretical final score varies with different configurations of the plane, such as the number of passengers, cargo and battery capacity. The key factor among all inputs is the number of passenger and cargo, which is directly related to the score in the ground mission and mission 2. Additionally, it is positively related to the size and battery capacity required by the plane, which contributes to score calculation in mission 2 and mission 3. At the end, the design target of the plane is set based on the results of the sensitivity analysis.

5.2 Methodology

To analyse how the score varies with different number of passengers (N_p) and cargo (N_c), we shall find the maximum possible score for different (N_p) and (N_c) combinations. However, there are numerous possible combinations, and it is unfeasible to calculate all of them. Therefore, we narrow down the scope of analysis to 16 combinations as follows:

Combination	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
N_p	10	10	10	10	20	20	20	20	30	30	30	30	40	40	40	40
N_c	3	6	9	12	3	6	9	12	3	6	9	12	3	6	9	12

Table 7 Sensitivity analysis combinations

The minimum N_p is set to 10 as preliminary score calculations suggests that it is more beneficial to carry more passenger and cargo. The minimum N_c is based on minimum N_p as the length of fuselage that can fit 10 passengers can fit around 3 cargo. The upper limit of N_p being studied is based on the length of fuselage. Minimum fuselage length to carry 40 passengers is around 1.5m (calculations explained in latter part of report). With the wingspan of aircraft limited to 1.5m by game rule, the wingspan to fuselage ratio is 1:1. Further extending the fuselage will lead to insufficient lift and poor handling due to limited wing area. The maximum N_c studied is 12 since the cargo is heavy and further increase in the number of cargo may require wingspan larger than 1.5m.

5.2.1 Battery capacity selection

To investigate the scoring of different combinations, the first step was to identify the battery capacity allowed on the plane. According to section 4.2.3 of the rule book, the propulsion battery cannot exceed 100 W-hrs in capacity. Considering that we are using 22.2V batteries, the maximum capacity is 4500 mAh.

Batteries of smaller capacities also need to be considered since some flight scenario may not require as much as 4500 mAh of battery capacity, and a smaller battery would make the plane lighter and fly at a even lower energy consumption rate. After reviewing market options, it was discovered that market available battery capacities of 22.2V batteries include 3300 mAh, 4200 mAh and 4500 mAh. During the score analysis for different passenger and cargo combinations, the usage of batteries with those capacities were considered and compared when determining the cruise speed.

5.2.2 Power requirement analysis

In this step, we set a designed cruise speed and calculates the power required to fly at such speed. We begin by referencing previous designs in the 2020 AIAA competition, which's topic is similar to that of this year. In 2020, the top 3 teams have drastically different design cruise speed. While the plane of University Nevada Las Vegas flew at around 21 meter per second, the design from University of Southern California flew at 42 m/s. Therefore, we decided to design the cruise speed at around 30 m/s, which provides room for increase or decrease based on testing results.

To calculate the power required at cruise, we begin by finding the loaded mass of the aircraft, which is the sum of empty mass of the plane and the payload. For the empty mass of the plane, we assume that the empty mass of the plane is 1.5kg when length is 0.8¹, and mass increase by 100g per 10 cm increase in length. Next, we set the wingspan and aspect ratio by referring to previous DBF aircraft design. Such values are then used to calculate the lift coefficient, induced drag coefficient and induced drag using the following equations:

¹ Length of plane is a function of number of rows of passenger. Assuming a minimum length of 0.4m due to electronics/avionics packaging at the nose and the tail of aircraft, the length of the fuselage increase by 7cm per row of passengers on top of the minimum length

Coefficient of lift C_l :

$$C_l = \frac{2Mg}{\rho S V^2}$$

Where

- M: Loaded mass of plane
- G: Gravity constant
- ρ : Air density
- S: wing area
- V: Cruise speed

Coefficient of induced drag C_{di} :

$$C_{di} = \frac{C_l^2}{\pi * AR * e}$$

Where

- C_l : Lift coefficient
- AR: aspect ratio
- e: efficiency factor

Induced drag

$$D_i = 0.5\rho C_{di} S V^2$$

Where

- ρ : Air density
- S: wing area
- V: Cruise speed
- C_{di} : Coefficient of induced drag

Due to uncertainty towards the CFD results and data related to interference drag. We model the total drag acting on the plane to be 1.5 times to that of the induced drag based on the ratio of induced drag coefficient to total drag coefficient of Cessna 172, which is 2.

Last but not least, we determine the power using the equation $P = DV$, where D is the total drag acting on the plane at cruise speed and V is the cruise speed.

Aircraft parameters	Symbol	Unit	Equation	Value
Designed Cruise speed	V	m/s	/	31
Passenger per row	R1	/	/	3
Number of rows	R2	/	$R2 = P // R1$	4
Min Cabin width	Wmin	m	$Wmin = R1 * 0.0635$	0.1905
Plane width	W	m	/	0.2
Min Cabin length	Lmin	m	$Lmin = R2 * 0.0762$	0.7048
Cabin length	L	m	/	0.8
Empty mass	Me	kg	$Me = 1.5 + 0.1 * (L - 0.8)$	1.5

Loaded mass	M	kg	$M = M_e \text{ or } M_e + M_{pc} \text{ or } M_e + M_b$	2.19
Wing span	S	m	/	1.2
Aspect ratio	Ar	/	/	6
Wing area	A _w	m ²	$A_w = S * S/Ar$	0.24
Lift coefficient	C _l	/	$C_l = 2 * M * 9.81 * / (1.224 * A_w * V^2)$	0.152204352
Drag coefficient	C _d	/	$C_d = C_{d0} + k*(C_l^2)$	0.081675913
Design Lift	Lift	F	Lift at cruise speed = Weight	21.4839
induced drag	D _w	F	$D_w = 0.5 * 1.224 * V^2 * A_w * C_d$	11.5286924
Estimated total drag	D	F	$D = 1.5 * D_w$	17.2930386
Power required	P	W	$P = D * V$	536.0841967

Table 8 Aircraft parameters related to power consumption calculations

5.2.3 Flight path analysis

To find out how many laps the plane can fly at the designed cruise speed, we break down the lap into different segments and determine the time taken and the horizontal displacement of the plane for different segments.

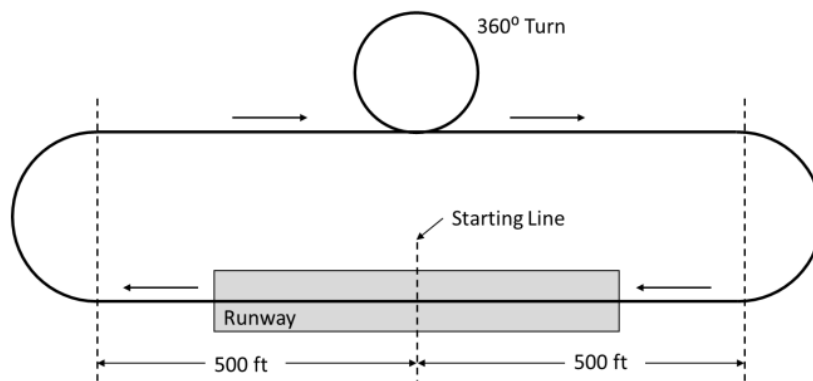


Figure 2 Figure of competition flight path

According to the rule book, the flight path can be broken down into 4 semi-circles sections and 2 straight sections. For the straight section, the horizontal displacement is 1000 ft and time taken to fly is found using $s = vt$. For the semi-circle sections, we assume the plane undergoes 2g of lateral acceleration and maintains altitude and cruise speed. We can calculate the horizontal distance travel using the equation $s = \frac{\pi V^2}{a}$ and find the time taken using the same equation above. With such information, we can calculate the lap length and lap time of the plane at cruise speed.

Assumptions	unit	value
Acceleration (take-off roll)	m/s ²	4
Acceleration (cruise)	m/s ²	2
Climb speed	m/s	20
Climb rate	m/s	5.176381
Cruise altitude	m	50
max lift coefficient	/	1.2
Stall speed	m/s	11.04039
G limit	g	3

Table 9 Assumptions made towards flight

Lap segment	Horizontal Length (m)	Average Speed (m/s)	Acceleration (m/s ²)	Time taken (s)
U-Turn	58.55194641	19.12252483	19.62	3.061935959
Straight cruise	304.8	31	0	9.832258065

Table 10 Flight segment information (cruise)

The take-off sequence was modelled by breaking it down into 3 parts. First, the plane is assumed to accelerate at 4 m/s² and takes off in 5 seconds. Then, it is models to climb on a slope of 15 degrees above ground at constant speed at 20 m/s to 50 m in altitude. After reaching cruise altitude, it is going to accelerate at 2 m/s² to cruise speed. The time taken and horizontal displacement of the entire take-off sequence is calculated using basic dynamic equations.

Lap segment	Horizontal Length (m)	Average Speed (m/s)	Acceleration (m/s ²)	Time taken (s)
Takeoff roll	50	10	4	5
Climb	186.6025404	19.31851653	0	9.659258263
Acceleration to cruise speed	140.25	25.5	2	5.5
Take off sequence	376.8525404	18.69377015	0	20.15925826

Table 11 Flight segment information (takeoff)

With information regarding lap time and time taken for takeoff, we can determine the number of laps that our plane can complete within 5 minutes.

Lap Parameter	Symbol	unit	Equation	Value
Lap length	LL	m	LL = 2*straight course + 4*U turn	843.8077857
Lap time	LT	s	LT = 2*straight course rime + 4*U turn time	31.91225997
Number of laps possible (take off considered)	L	/	L = Takeoff sequence length/Lap length + (300 - takeoff sequence time)/lap time	9.215676335

Table 12 Lap time and possible number of laps

5.2.4 Feasibility validation

After calculating the number of laps possible with the design flying speed, we need to validate whether flying at such speed is feasible with the limited battery capacity. The battery life is calculated by dividing the battery capacity (in watt hour) by the power consumption of during cruise.

$$FTA = \frac{Cap}{60} * P_a$$

Where

- FTA: Flight time available
- Cap: Battery capacity in mAh
- Pa: Average power consumption per lap²

To provide buffer, FTA should not below 6 minutes such that the plane can return to land safely after the mission. If the FTA is below 6 minutes, the designed speed is decreased. However, if the lap count is decreased due to a decrease in design speed, a battery with a larger capacity is considered to maintain the number of laps available since this variable significantly affect the number of points in mission 2 and mission 3.

5.2.5 Mission 3 analysis

For mission 3, it was assumed that the plane fly the same configuration (motor and battery) as mission 2. However, there are 2 dependent variables that make the sensitivity analysis challenging. Both banner size and laps flown significantly affects the score but they are negatively related.

² Note that average power consumption per lap is higher than the power required to maintain cruise as more power is required during steep turns. The average power consumption is calculated by dividing the energy consumed during a lap by the lap time.

To simplify the process, it was decided to fix the number of laps and maximize the banner size for each case.

Banner parameters	Symbol	Unit	Equation	Value
Length	Lb	m	/	2.5
Aspect ratio	Ar	/	/	5
Air Density	d	kg m ⁻²	/	1.224
Mass	Mb	kg	$Mb = Lb * Lb / Ar * d$	0.15
Drag coefficient	Dcb	/	/	0.03
Drag	Db	N	$Db = 0.5 * Dcb * d * Lb * Lb / Ar * V^2$	22.07296875

Table 13 Banner parameters calculation

The banner drag is calculated using the following equation and taken into account when calculating the total drag of plane and power consumption of flight during cruise.

$$D_b = 0.5 * D_{cb} * \rho * S_b * V^2$$

Where

- D_b : Drag of banner
- D_{cb} : Drag coefficient of banner
- ρ : Air density
- S_b : Banner area
- V : Designed cruise speed

5.2.6 Score calculation for each combination

Finally, the maximum score possible for each combination is calculated and compared

5.3 Results and discussion

5.3.1 Score results

Ground mission																
Mission time equation: $2 * 3 * (\text{number of passengers} + \text{number of cargo})$																
Case	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Mission Time	78	96	114	132	138	156	174	192	198	216	234	252	258	276	294	312
Score	1	0.8125	0.68421	0.59091	0.56522	0.5	0.44828	0.40625	0.39394	0.36111	0.33333	0.30952	0.30233	0.28261	0.26531	0.25

Table 14 Scores for ground mission

Mission 2																
Net income = $P * (6 + 2L) + C * (10 + 8L) - L * EF * (10 + 0.5P * 2C)$																
Case	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
No of P	10	10	10	10	20	20	20	20	30	30	30	30	40	40	40	40
No of C	3	6	9	12	3	6	9	12	3	6	9	12	3	6	9	12
Cruise speed	32	32			31	34	34	33	32	31	29	29	29	30	31	31
Motor selected	x3120	x3120			x3120	x4120	x4120	x4120		x4120	x4120	x4120	x4120	x4120	x4120	x4120
Battery capacity	3300 mah	3300 mah			3300 mah	4200 mah	4200 mah	4200 mah		4200 mah	4200 mah	4200 mah	4200 mah	4200 mah	4500 mah	4500 mah
No of laps	9	8			9	9	9	8		9	8	8	9	9	9	8
Cap (Wh)	73.26	73.26			73.26	93.24	93.24	93.24		93.24	73.26	73.26	93.24	93.24	99.9	99.9
EF	0.7326	0.7326			0.7326	0.9324	0.9324	0.9324		0.9324	0.7326	0.7326	0.9324	0.9324	0.999	0.999
Net income	307.539	465.758			474.572	623.469	819.119	919.795		781.511	953.986	1140.82	743.902	939.553	1106.43	1176.43
Score	1.26142	1.39591			1.4034	1.52997	1.69627	1.78185		1.66431	1.81091	1.96973	1.63234	1.79865	1.9405	2

Table 15 Scores for mission 2

Mission 3																
Assume flying in same configuration as mission 2																
Case	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Number of laps	9	9			9	9	9	9		9	9	9	9	9	9	9
Banner length (m)	1.7	1.7			1.7	2.3	2.3	2.3		2.5	2.5	2.5	2.3	2.3	2.5	2.5
Wingspan	1.2	1.2			1.3	1.3	1.3	1.3		1.4	1.4	1.4	1.5	1.5	1.5	1.5
RAC	0.822	0.822			0.828	0.828	0.828	0.828		0.834	0.834	0.834	0.84	0.84	0.84	0.84
Cache	18.6131	18.6131			18.4783	25	25	25		26.9784	26.9784	26.9784	24.6429	24.6429	26.7857	26.7857
Score	2.68993	2.68993			2.68493	2.92667	2.92667	2.92667		3	3	3	2.92	2.92	3	3

Table 16 Scores for mission 3

Case	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Total score	5.95134	5.89833			5.65354	5.95663	6.07122	6.11477		6.02542	6.14425	6.27925	5.85466	6.00125	6.2058	6.25

Table 17 total score

Note that for mission 1, teams are awarded 1 point upon completion and some cases are omitted due to design feasibility.

5.3.2 Result analysis

Among the combinations studied, case 12 scored the highest, with 6.28 points out of 7. Case 5 score the lowest, with only 5.65 points.

Generally, the score is positively related to the passenger and cargo capacity of the plane. However, an outlier is case 1, which scores 3.95 points, only 0.32 points behind the highest score. This is mostly contributed to the high score in the ground mission.

5.3.3 Conclusion and design goal setting

In conclusion, to maximize point scoring, teams shall design a plane to load as much passenger and cargo as possible under the allowable wingspan. However, for this project, it was decided to build a plane that could only carry 10 passengers and 3 cargo (combination 1 for sensitivity analysis). The reasons are as follows:

- The score difference between combination 1 and the highest scoring combination is just 0.32. Due to assumption and feasibility limitations in the sensitivity analysis, it is necessary to conduct flight tests with different prototypes to validate the assumptions made and the analysis model. The plane with the first passenger and cargo is ideal for this purpose as its size is drastically different from the highest scoring combination yet the theoretical points for the 2 combinations is very similar.
- The HKU DBF team is already building a prototype that could carry 48 passengers, therefore, the team can develop a plane in the opposite direction by developing a small plane to make effective comparison during testing.
- There is only 2 months for the group to develop the prototype due to course schedule. It would be unfeasible for the team to build a large plane with wingspan exceeding 1.2 meters due to time and workspace constraints.

6. Preliminary Design (Prepared by Zhao Zehui)

6.1 Design Methodology

The team's design methodology built on previous HKU DBF techniques but also incorporated new techniques learned from coursework, off-season research, and industry experience. The importance and the weighing of each mission contributing to the optimization of the final performance is determined by sensitivity analysis. From this, it is determined that selecting a certain amount of payload and maximizing the possible banner length would contribute to the best score result. From this, a constraint diagram was then constructed based on these requirements to further narrow the design choices into power and wing loading. Aerodynamic analysis started with airfoil selection and geometric parameters of the wing, tail and fuselage,

simulated in XFLR5 and then ANSYS. Stability was then obtained from XFLR5 and compared with hands-made calculation and real-life testing. Trade studies were conducted to optimise for the configuration that produced the best score under the stated requirements. If the preliminary design met all the system design requirements, the best scoring design was taken. A CAD model was produced to meet the requirements for the payload anticipated, which then manufactured through laser-cutting and 3D printing.

6.2 Aerodynamic Analysis

Power estimation for the aircraft:

The power of propeller driven aircraft in steady and level flight can be calculated using

$$D = \left(\frac{1}{2} \rho S C_{D0}\right) V^2 + \left(\frac{2kW^2}{\rho S}\right) V^{-2} \quad (1) \text{ and } P = \frac{DV}{\eta} \quad (2)$$

From previous section, it was knowing the MTOW would be initially estimated as 3 kg. The expected payload weight was 10 passengers with 3 cargos, in a total of 0.71 kg. With this data, the main wing parameters calculation, airfoil selection and stability analysis can be conducted.

Basic parameters for the aircraft main wing:

Wing area	Span	Chord	Wing loading	Aspect ratio
0.24 m ²	1.2 m	0.2 m	12.5	6

Table 18 Basic Parameters for the Main Wing

Since the range of the wingspan must fall in the range of 3 foot to 5 foot, with consideration of the length of the aircraft and the amount of payload, the span will be 1.31 m, with an aspect ratio of 6. This configuration aims to increase the aircraft speed. A rectangular wing is selected, as the aircraft is designed to fly in subsonic speed, and the CG is in an appropriate position. [Gudmundsson]

	Aileron (in mm)	Flap (in mm)	Wing tip (in mm)
Length	350	160	15
Height	40	30	200

Table 19 Specifications for Aileron, Flaps and Wing Tip

6.2.1 Airfoil selection

Renolds numbers ranging from 200,000 and 2 million were simulated for each airfoil in XFLR5, with the result exported into an excel which was weighted based on the main requirements set out in the constraint diagram. This consisted of the maximum lift coefficient, lift to drag in cruise/turn and the pitching moment. Around 10 different airfoil were simulated, from these the top three best performing airfoils were taken into 3D space, with data exported and simulated using the mission model, which calculated the score on M2 and M3. This process resulted in top performing airfoils namely ClarkY, Davissm, and Wortman FX. The team then investigated blending airfoils using XFLR's interpolate tool using 65 % of Clark-Y and 35% of WFX, which allows for the team to have the high lift-to drag from the WFX and also a better stall sensitivity from Clark-Y.

Here are the analysis results of all the airfoils in terms of C_L , C_D , X_{tr} , and C_m .

C_L	Coefficient of Lift, which affects the lift force induced by the main wing.
C_D	Coefficient of Drag, which affects the drag force induced by the main wing.
Xtr	Flow transition point, where the laminar flow turns to turbulent flow on the upper surface of the main wing.
C_m	Coefficient of moment, which affects the moment induced by the main wing.

Table 20 Meaning of Different Parameters

Colours and symbols of different airfoils:

NACA 0009 - T1_Re0.300_M0.00_N9.0 - T1_Re0.500_M0.00_N9.0 - T1_Re1.000_M0.00_N9.0					
NACA 0012 - T1_Re0.300_M0.00_N9.0 - T1_Re0.500_M0.00_N9.0 - T1_Re1.000_M0.00_N9.0			CLARK Y AIRFOIL - T1_Re0.300_M0.00_N9.0 - T1_Re0.500_M0.00_N9.0 - T1_Re1.000_M0.00_N9.0		
			Davisssm airfoil - T1_Re0.300_M0.00_N9.0 - T1_Re0.500_M0.00_N9.0 - T1_Re1.000_M0.00_N9.0		
SD7037 AIRFOIL - T1_Re0.300_M0.00_N9.0 - T1_Re0.500_M0.00_N9.0 - T1_Re1.000_M0.00_N9.0			MH 60 AIRFOIL - T1_Re0.300_M0.00_N9.0 - T1_Re0.500_M0.00_N9.0 - T1_Re1.000_M0.00_N9.0		
			WORTMANN FX 63-137 AIRFOIL - T1_Re0.300_M0.00_N9.0 - T1_Re0.500_M0.00_N9.0 - T1_Re1.000_M0.00_N9.0		

Table 21 Airfoils List

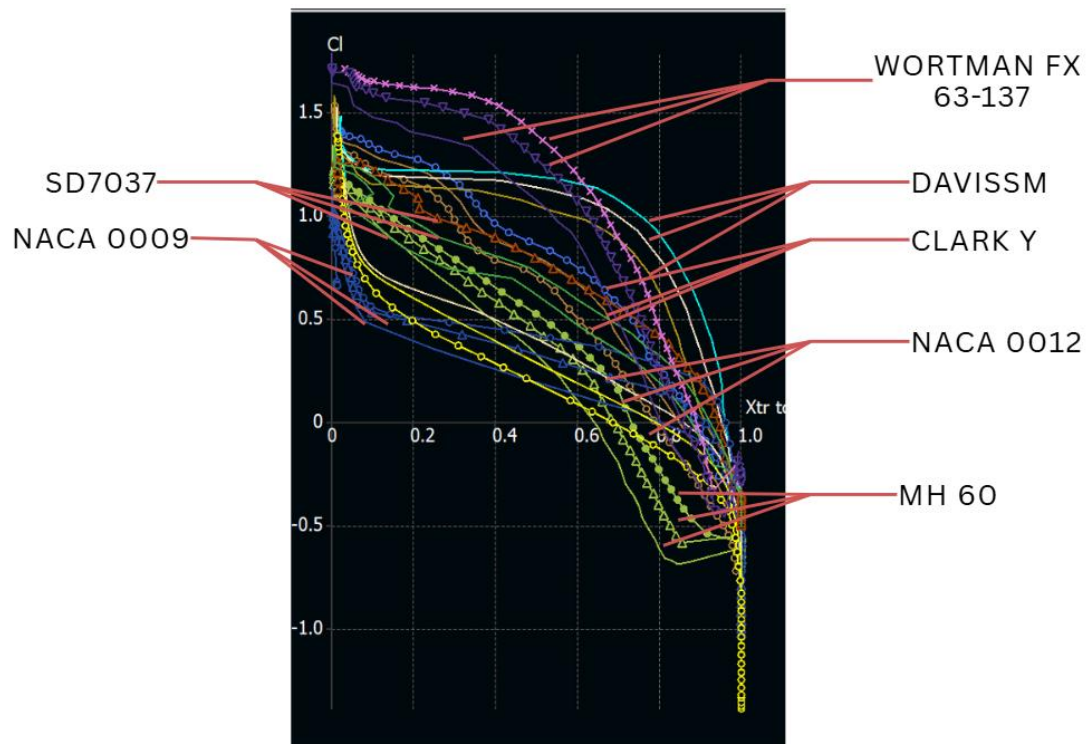


Figure 3 C_L VS Xtr

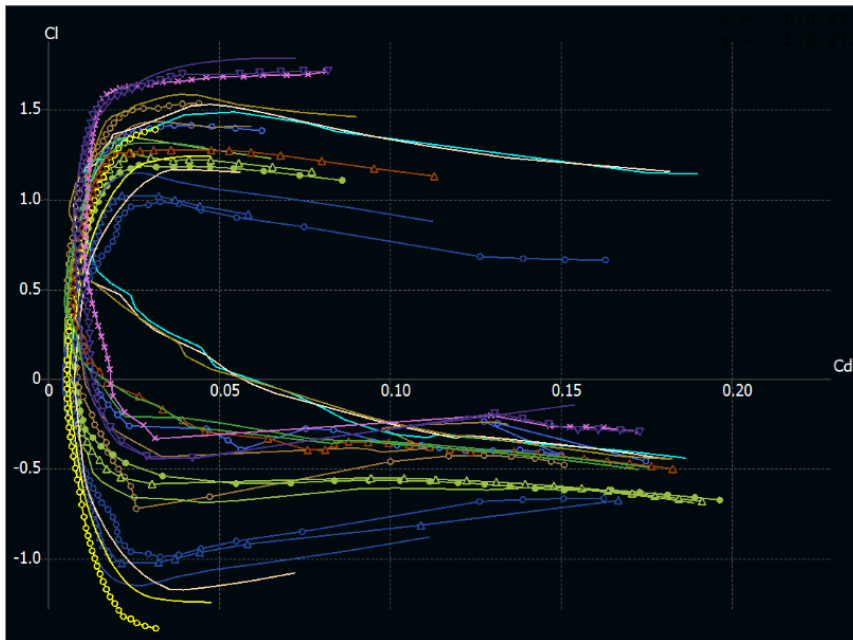


Figure 4 C_L VS C_D

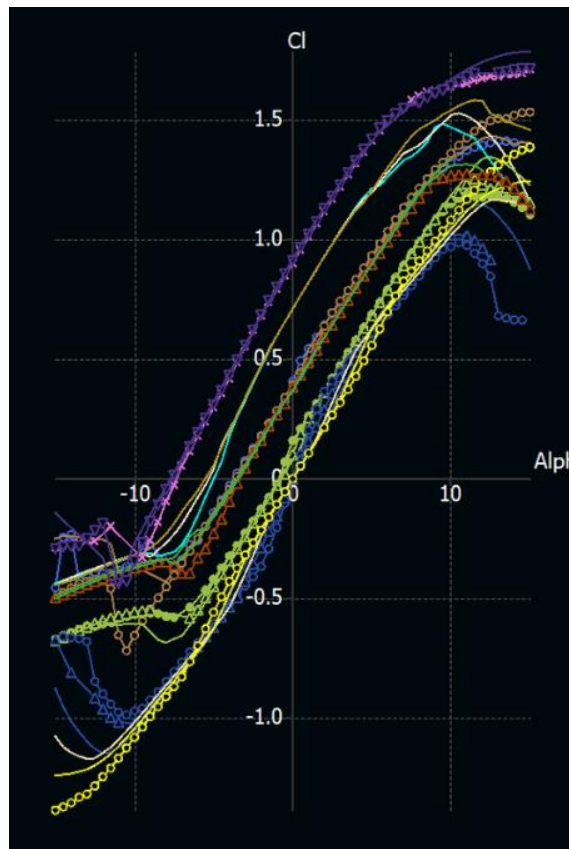


Figure 5 C_L VS AoA

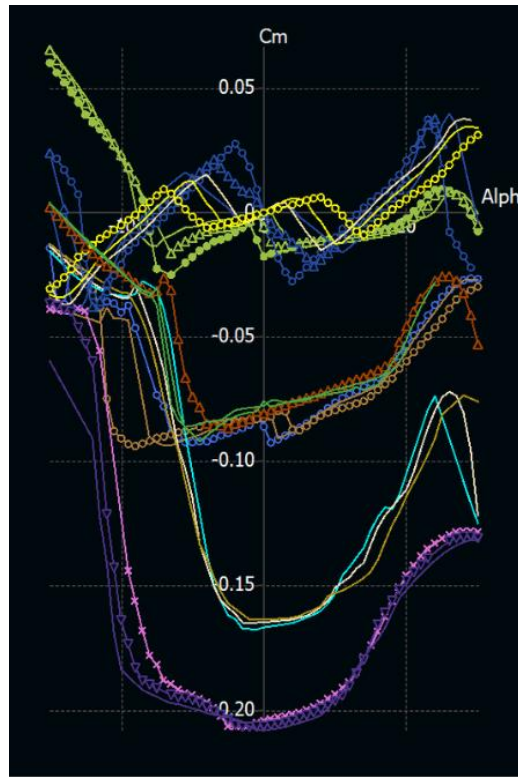


Figure 6 C_m VS AoA

The results of these graphs conclude the best performing airfoils are WFX and Clark-Y airfoil. Hence, a hybrid of these two airfoils is selected.

6.2.2 Wing twist and dihedral angle

The team investigated the use of twist to increase the performance of the aircraft and alignment of the two blended airfoils. The twist angle is applied on the outboard region where Clark-Y airfoil is adopted. The dihedral angle is 1 degree from the root chord of the wing, as larger degree will result in extrusion of the wing rod. The result is simulated in XFLR5 and concluded that no washin or washout will be adopted for the wing design.

AoA	Dihedral angle	Wing twist angle	C_L	C_D	Efficiency	C_P	C_m
0	1	0	0.374	0.025	0.727	0.75	-0.015
0	1	-0.5	0.357	0.024	0.684	0.76	-0.014
0	1	-2	0.306	0.024	0.528	0.7	-0.01
6	1	0	0.859	0.059	1.025	0.78	-0.514
6	1	-0.5	0.910	0.061	1.011	0.86	-0.514
6	1	-2	0.859	0.059	0.941	1.05	-0.512

Table 22 Wing Twist and Dihedral Angle Comparison

6.2.3. Drag analysis

To size the propulsion system, the drag in cruise was required. Further, the team identified drag as a crucial factor for this competition due to the battery limit set out in the rules. From this, the team conducted a preliminary drag analysis on all components using the analysis method from General Aviation Aircraft Design by Snorri Gudmundsson and simulation on ANSYS for banner drag which will be introduced in section 7.4. This analysis involved calculating the skin friction drag of all wing-like elements and parasitic drag along the wing surfaces, using the formula (1).

6.2.4 Stability Analysis

The stability analysis conducted on Xflr5 regarding the performance of the aircraft on the ratio of C_m VS AoA, C_L/C_D vs AoA, C_p vs AoA and C_L VS AoA are shown in the following graphs. Two types of tail were considered during the analysis: T tail and conventional tail, which are named as T-design and C-design. From the analysis results, it is shown that T tail generates a larger coefficient of lift and coefficient of drag than the conventional tail, though the conventional tail design has a higher C_L/C_D than that of T-tail.

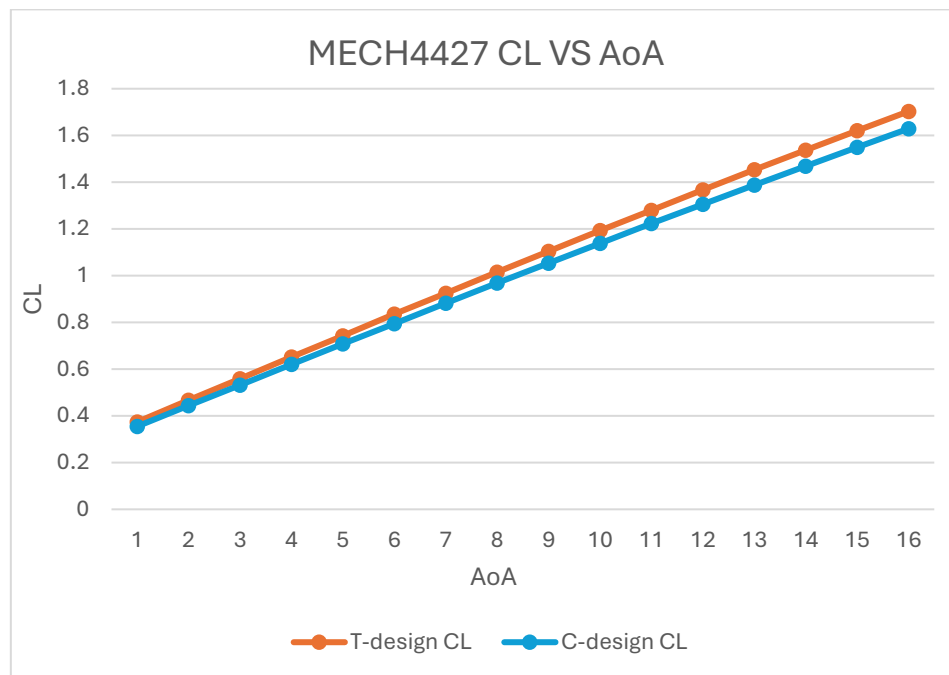


Figure 7 C_L vs AoA for T-design and C-design

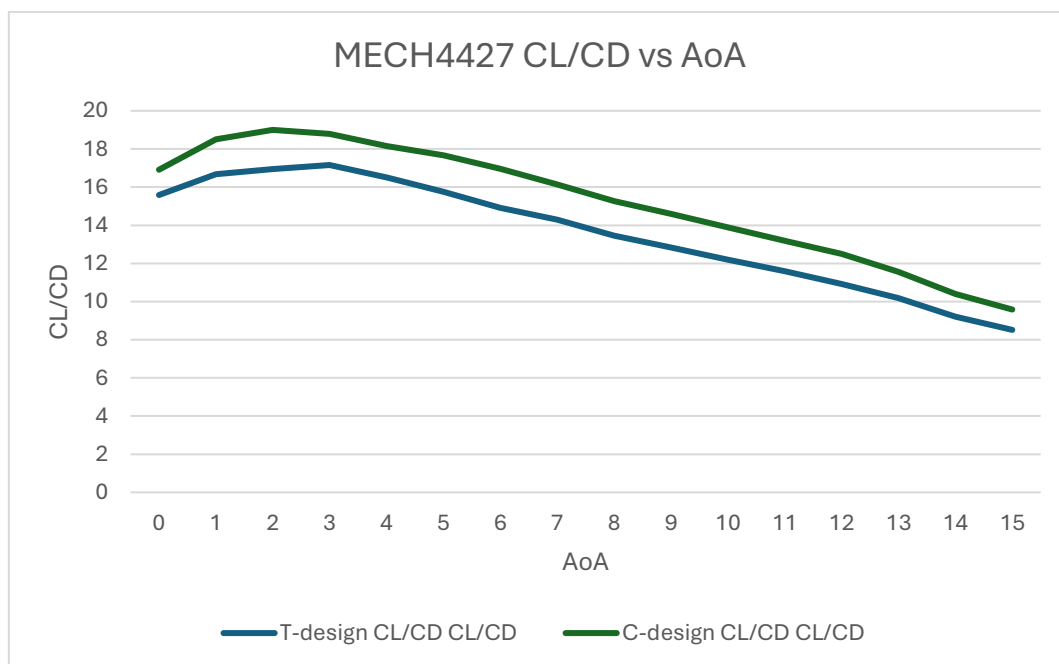


Figure 8 , C_L/C_D vs AoA for T-design and C-design

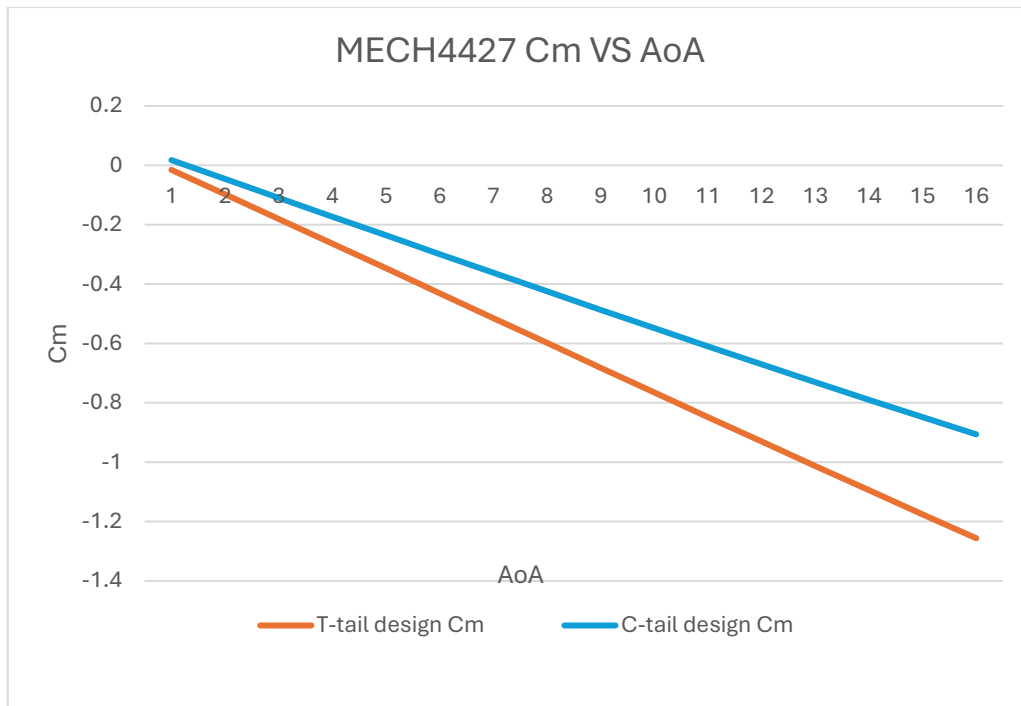


Figure 9 C_m vs AoA for T-design and C-design

Although the overall coefficient of moment induced by the C-design is smaller than that of T-design, the coefficient of moment of C-design for the first few degrees has an opposite sign than the value coefficient of moment values in higher degrees, which gives more challenge for the pilot to adapt the sudden change of direction of the force they feel when flying the aircraft, as the AoA goes up.

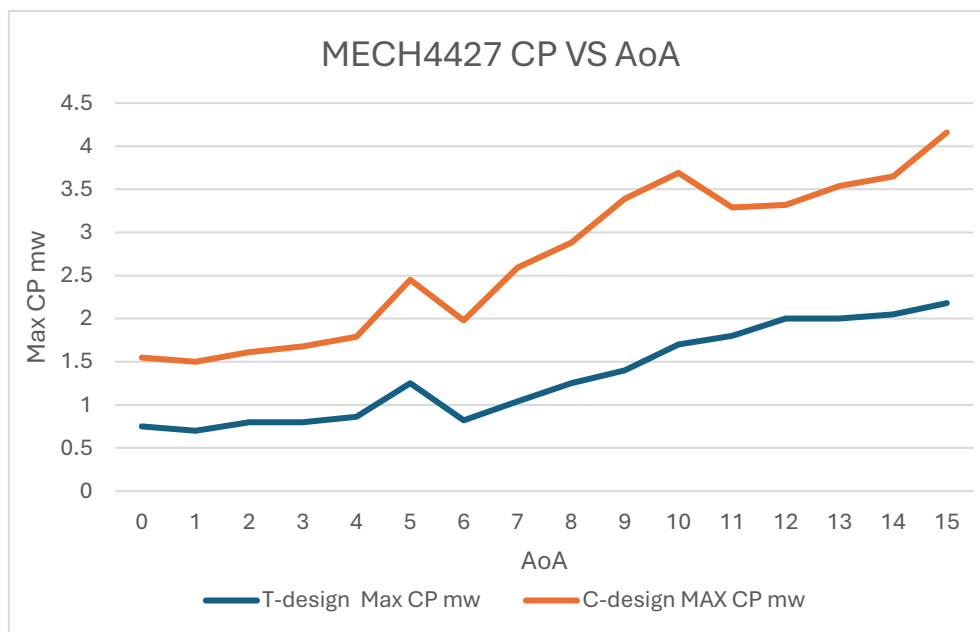


Figure 10 Max C_p mw vs AoA T-design vs C-design

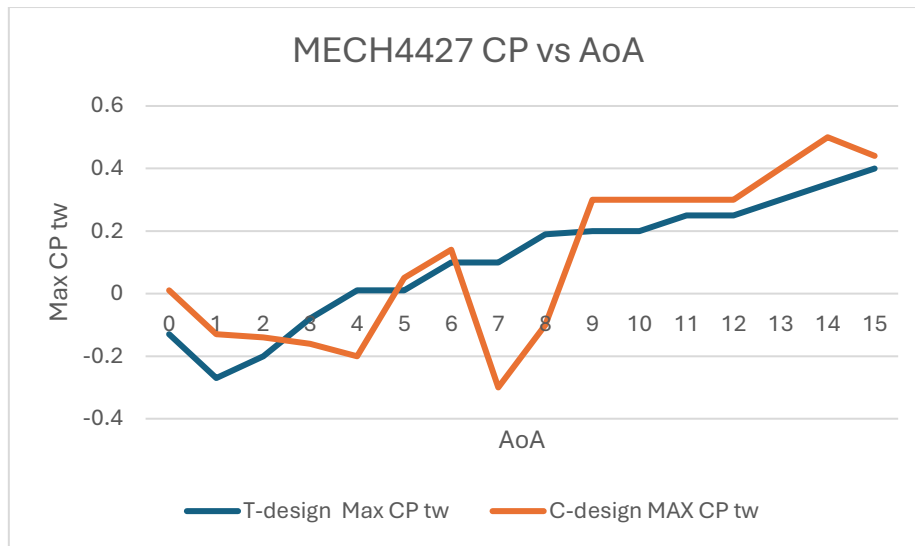


Figure 11 Max C_p tw vs AoA T-design vs C-design

Furthermore, the coefficient of moment of the T-design is 0.35 larger than C-design near stall, but the coefficient of pressure inserted on the main wing and tail wing in C-design is 2.0 larger than that of T-design, showing that a larger pressure force needs to be endured by the aircraft, giving more challenges to the aircraft structural rigidity and hence affects the manoeuvrability. The term mw refers to main wing and the tw refers to tail wing. Through trade-off consideration, the group weighs ease of control, force endurance and lift generation as more important criteria. T-tail design is adopted, and the detail of the aircraft design is discussed in later sections.

6.3 Battery Selection

Based on the power required by the aircraft to achieve desirable outcome, the choice of a single motor configuration is adopted. Several brands of motors are analysed and compared. A six-cell (22.2 V) Sunnysky 3000 mAh 30 C LiPo battery was selected to optimize the weight and the battery capacity for this aircraft. The packs would have an allowable maximum discharge of 65 A, which falls within the maximum current limit of 100 A.

6.4 Motor Selection (Prepared by Zhao Zehui & Zhong Wendi)

Several scenarios are considered during the motor selection process. Four brands of motors are selection for comparison, include Scorpion series, Sunnysky series, Dualsky series and T-motor series. The first scenario assumes carrying the largest amount of payload, which requires a motor thrust of around 7000 to 8000g. Parameters include the weight, Kv (rpm), maximum thrust, maximum current, peak power, voltage, propeller size, temperature and price of the motor are considered.

	Weight	KV	Max Thrust	Max Current	Peak Power	Voltage	Propeller Size	Temperature	Price in USD
Scorpion A-4220-540kv V2	288	540	7000g	115	2553	22.2	16*8	-	199.99
Sunnysky X4120 V3	292	480	6780g	97	1793.76	22.2	16*8	130	105.99
Dualsky XM5060 EA-10	392	490	8308g	110.2	2280	22.2	16*10	-	112.06
T-motor AT4125 Long Shaft	355	540	8492g	85	1718.7	22.2	16*8	97	115.99

Table 23 Motor Selection for high thrust

From the comparison table, it is shown that the Scorpion X4120 provides a decent amount of thrust while being the lightest motor of all. Though, its maximum current is 115, which exceeds the current limitation from the AIAA DBF rules, it is acceptable if the battery only draws a maximum of 100A to the motor. In terms of light-weight motor consideration, the second choice would be SunnySky X4120, as it has the second lightest weight out of the four motor brands. However, these two motors are only able to generate a thrust of 6780g to 7000g, which is suitable for aircraft lighter than such values. In terms of thrust generation, T-motor AT4125 would be the choice of all, as it has the largest thrust generation and lighter than Dualsky XM5060 at the same time. Though, the maximum current that can be drawn to the motor is 85A, it can reduce the efficiency factor, hence reduce the chartered flight cost in mission 2.

The second scenario assume carrying 10 passengers and 3 cargo which is the current selected scenario for the group. Motors between 3000g to 5045g are considered. The group considered the required power, weight of the motor and necessary thrust for the aircraft. Since the aircraft weights 3kg, the SunnySky series X3120 is selected.

	Weight	Kv	Max Thrust	Max Current	Peak Power	Voltage	Propeller Size	Temperature	Price in USD
Scorpion A-3017	136g (154g mounted)	600	5030g	43	962	22.2	14*6	-	129.99
Sunnysky X3520	219	560	4430g	80	1478.52	22.2	14*7	90 (1 min)	59.99
Dualsky XM4255	269	520	5045g	59.2	1344	22.2	14*7	-	72.7
T-motor AT3520	221 (with cables)	550	4116g	70	1100	22.2	14*7	59	79.99
SunnySky X Series V3 X3120	159	585	3560g	65	1625	22.2	12*6.5	130 (3 min)	47.99

Table 24 Motor Selection for Low Thrust

7. Detailed Design

7.1 Fuselage (Prepared by Zhong Wendi)

7.1.1 Section overview

This section reviews the design process of the fuselage for the airplane. The main process involves deciding on the structure, materials, and dimensions of the fuselage, design and analyse the fuselage structure on SolidWorks, design different compartments and components in the fuselage and manufacturing.

The table below shows the basic parameters of the fuselage

Parameter	Description/Value
Structure type	Truss structure
Material	Balsa wood
Dimensions	810mm*150mm*162mm
Number of passengers	10
Number of cargo	3

Table 25 Specifications of fuselage

7.1.2 Fuselage structure and material

The fuselage is the central body of a fixed-wing airplane, serving as the main structural component that houses the crew, passengers, cargo, and sometimes fuel or equipment. It connects the wings, empennage (tail assembly), and landing gear, while withstanding aerodynamic forces, pressurization (in larger aircraft), and various loads like bending, torsion, shear, and compression during flight. The design must balance strength, weight, aerodynamics, and manufacturability. Fuselage structures have evolved from early fabric-covered frames to advanced composite shells, influenced by materials like wood, metal (aluminium or steel), and composites (e.g., carbon fibre).

There are several primary structural options for fuselages, each with distinct construction methods. The most common types are truss, monocoque, and semi-monocoque.

Truss structure

This is a framework of interconnected beams, tubes, or struts (often made of wood, steel, or aluminum) forming a rigid lattice, typically covered with fabric, thin plywood, or metal sheets for enclosure. It's like a skeletal frame where longerons (longitudinal members) and struts handle primary loads, with diagonal bracing for stability.

On the one hand, this structure is lightweight for small aircraft, easy and inexpensive to construct and repair and provides good resistance to tension and compression loads by allowing for flexibility in shape. On the other hand, it may have poor aerodynamics due to non-streamlined shape and more limited internal space compared to other structures.

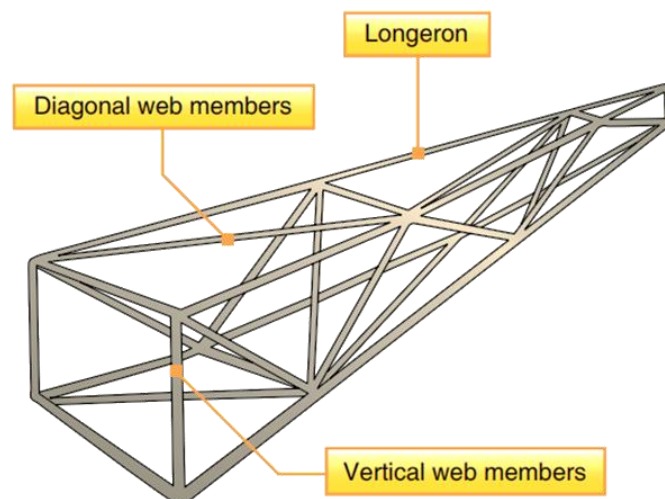


Figure 12 Truss structure

Monocoque structure

This structure is a "single-shell" design where the outer skin (often aluminium, plywood, or composites) bears nearly all structural loads without much internal bracing. It's like a hollow tube or eggshell, relying on the skin's thickness and curvature for strength against bending, torsion, and shear. Early versions used molded plywood; modern ones may incorporate minimal reinforcements.

Although it is highly aerodynamic with smooth surfaces reducing drag and lightweight and maximizes internal space, it is intolerant to surface deformation or damage and challenging to manufacture precisely.



Figure 13 Monocoque structure

Semi-monocoque structure

This is a hybrid where a thin outer skin is reinforced by an internal substructure of frames (bulkheads or formers), stringers (longitudinal stiffeners), and longerons. The skin handles shear loads, while the internals manage bending, compression, and torsion. This is the most common type in modern aircraft, often using aluminium or composites.

This design balances strength, rigidity, and low weight with excellent strength-to-weight ratio and prevents buckling through reinforcements by distributing loads efficiently. However, it is more complex to design and manufacture than truss types; potential for hidden damage in composites and precision assembly requires high initial cost.

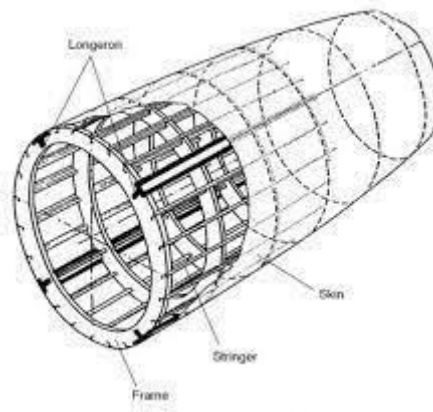


Figure 14 Semi monocoque structure

A decision matrix is used to determine the structure and material used in this project. 5 means the structure is more superior, 1 means the structure is inferior. Truss structure is selected since it scored the highest in the decision matrix.

Parameter	Weighting	Truss structure	Monocoque structure	Semi-monocoque structure
Ease of manufacturing	3	5	1	1
Cost	3	5	1	1
Weight to strength ratio	2	2	5	3
Health hazard	2	5	1	1
Ease of assembly	2	3	4	1
Ease of modification	1	3	2	2
Total score		53	28	18

Table 26 Decision matrix for fuselage structure

For material, balsa wood is selected for the following reasons:

- High strength to weight ratio. The specific strength of balsa wood is 521 kN m/kg, which is very high compared to aluminium (115 kN m/kg) and 3D printed PLA (48 kN m/kg).
- Low cost. Wood is relatively cheap compared to aluminium and carbon fibre, Balsa wood plates are free of charge in Innowing.
- Manufacturability Balsa wood can be cut using laser cutter in Innowing, which is much faster and convenient than the manufacturing processes of other materials such as carbon fibre, which involves waterjet or CNC for aluminium.

7.1.3 Dimensions design

The dimensions of the fuselage shall be minimized to reduce the form drag of the plane.

First, the width of the plane is determined based on the payload requirements, specifically the number of passengers per row. Each passenger is modelled to be 70mm in width and structure components are assumed to be 10mm in thickness. If the plane is designed to carry 1 passenger per row, the width of the plane cannot fit the cargo. If the plane carries 3 passengers per row, the fuselage will not have enough length to hold 3 cargos. Therefore, the plane is designed to carry 2 passengers per row and the width of the fuselage is set to be 150mm.

However, it is weight and aerodynamic inefficient to apply the width to the entire plane. Therefore, both the nose and the tail of the plane was narrowed. The width of the nose was set to 80mm to mount the motor and the tail of the plane was set to 40mm to mount the vertical stabilizer.

Next, the height of the fuselage is determined, which is based on the layout of the fuselage. Generally, the highest part of the fuselage should be capable of housing 1 cargo, 1 passenger and the thickness of the wing arranged vertically. After calculations, the highest part of the fuselage was set at 162mm. The nose and tail of the plane also narrowed to optimize aerodynamics and weight. The height of the nose is 80mm and the height of the tail is 80mm.

Finally, the length of the fuselage was determined by adding the minimum length to house the passengers, battery and avionics. Each passenger is 70mm in length, the fuselage need to carry 5 rows of passengers. The battery need a volume with length at least 150mm. Another 150mm is expected to house all avionics. Therefore, the absolute minimum length of the fuselage is 650mm. In the end, the fuselage length is set to be 810mm after including structure components and different mechanisms.

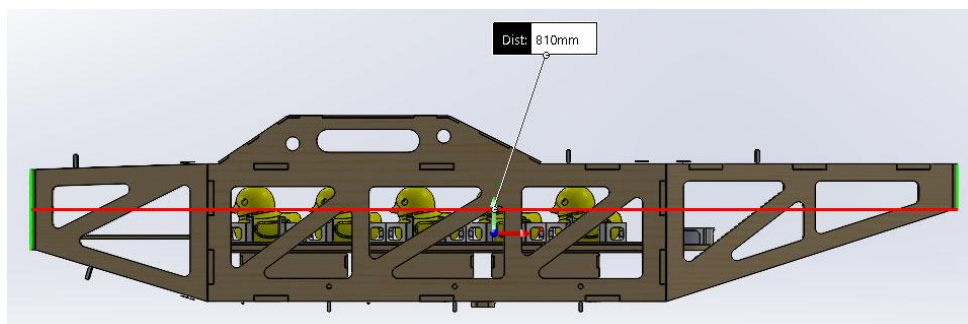


Figure 15 Fuselage maximum length

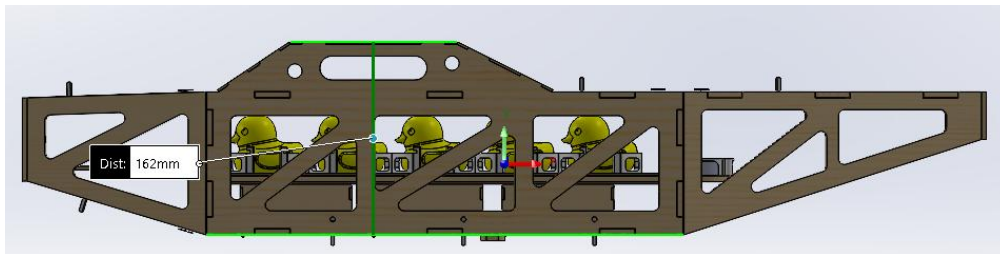


Figure 16 Fuselage maximum height

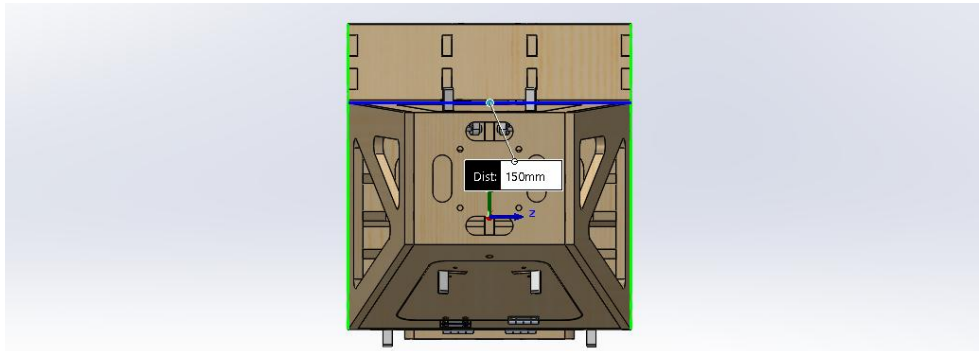


Figure 17 Fuselage maximum width

7.1.4 Structure design

There was multiple consideration when designing the structure of the plane.

- Structure triangulation

In a triangulated structure, beams, bars, or struts (called members) are arranged to form triangular units. These units distribute loads across multiple paths, reducing stress on individual components. For instance, in a truss (a common triangulated framework), joints (nodes) connect the members, and the structure relies on axial forces (push or pull along the member's length) rather than bending moments. Meanwhile, a square or rectangle can shear into a parallelogram under lateral forces, but adding a diagonal brace creates two triangles, locking the shape in place. This "triangulation" effect stabilizes the entire assembly.

- Bulkhead reinforcement

2 Bulkhead were added at the connection at the 2 sends of the centre compartment to reinforce the structure at the connection. Additionally, small triangles are added around the bulkhead to increase the gluing surfaces between the compartments.

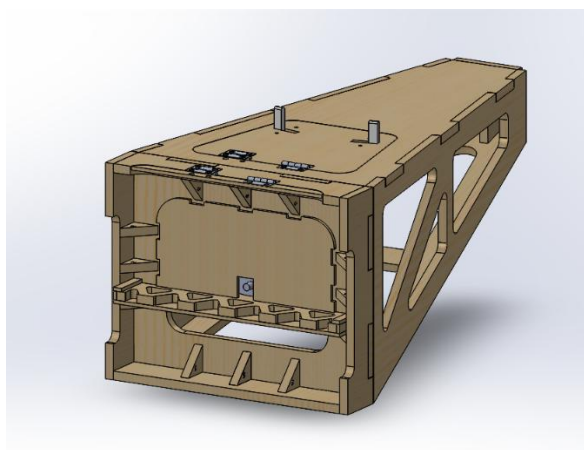


Figure 18 Bulkhead reinforcement

- Wing box reinforcement

For the wing mounting, a pair of additional 5mm balsa wood plates are added on the top of the fuselage to mount the wing. This structure reinforces the wing mounting, allowing the plane to do steep turns safely during mission 2 and 3, minimizing the lap time.

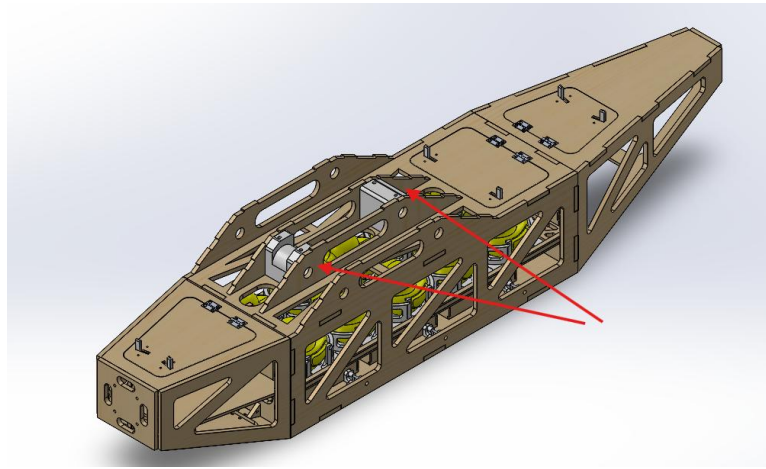


Figure 19 Wing box reinforcement

7.1.5 Strength analysis

After completing CAD of the basic fuselage structure, FEA is conducted to verify its strength under high loading during flight and during hard landing.

For simulating high loading conditions, we fix the geometry of the wing box and apply 3g of acceleration to the entire structure. Additionally, the weight of the motor, passenger and cargo to respective contact surfaces are scaled and applied. Furthermore, full motor thrust is applied on the motor mounting plate. Results show low stress loading on the structure components relative to the yield strength of balsa wood.

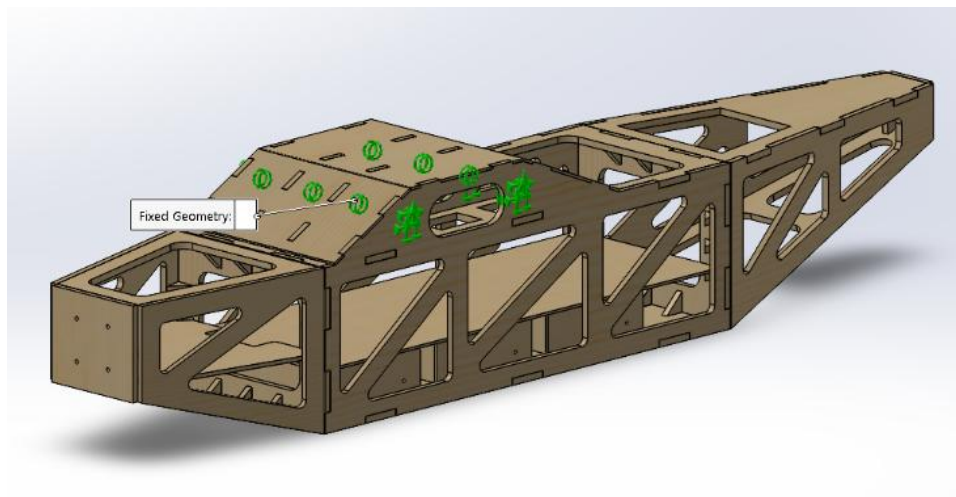


Figure 20 Location of fixed geometry

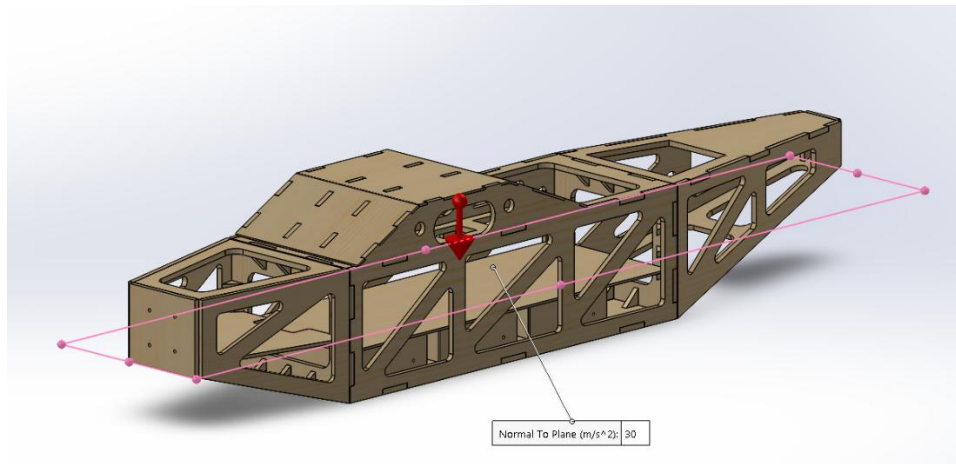


Figure 25 3g of acceleration load

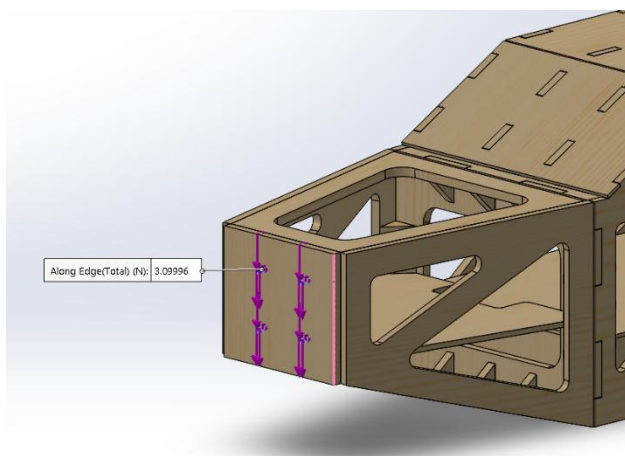


Figure 21 Motor weight

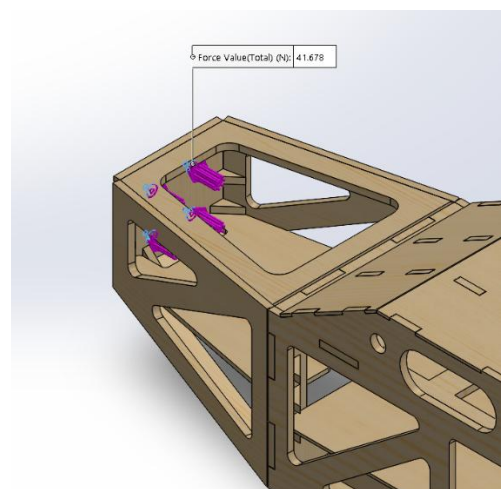


Figure 22 Motor thrust

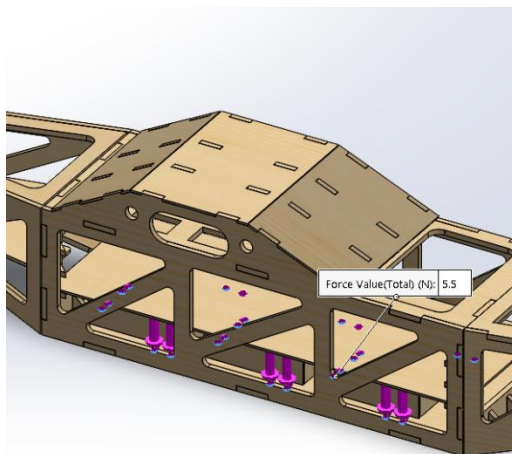


Figure 23 Cargo load

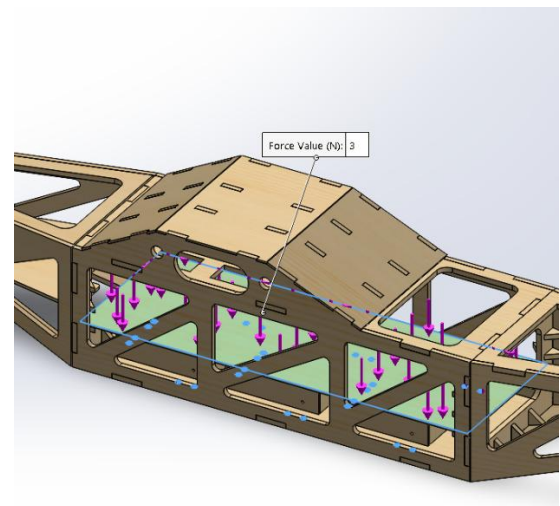


Figure 24 Passenger load

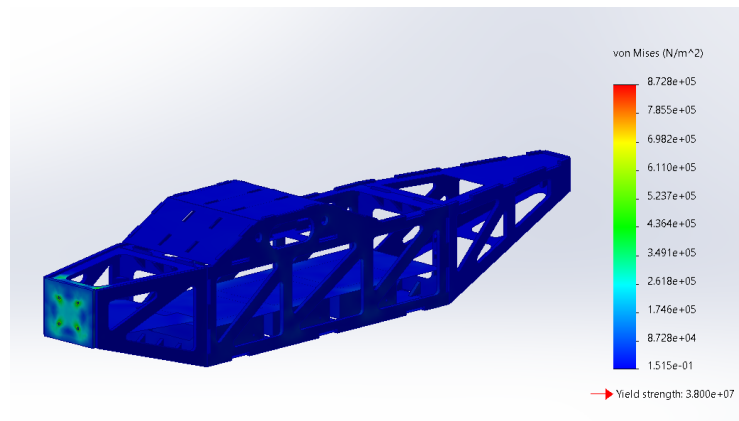


Figure 26 Stress on fuselage under 3g load

For simulating hard landing, the geometry is also fixed at the wing box. For loading, the acceleration load and the thrust of the motor are removed, however, a load equals 2 times the weight is applied to the landing gear mounting surfaces to simulate the load of a hard landing. Results show low stress loading on the structure components relative to the yield strength of balsa wood.

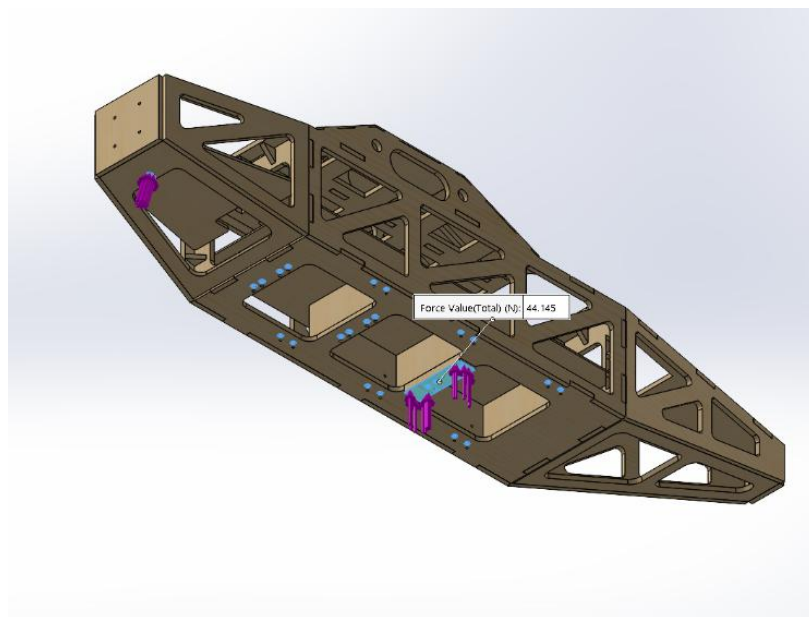


Figure 27 landing gear load

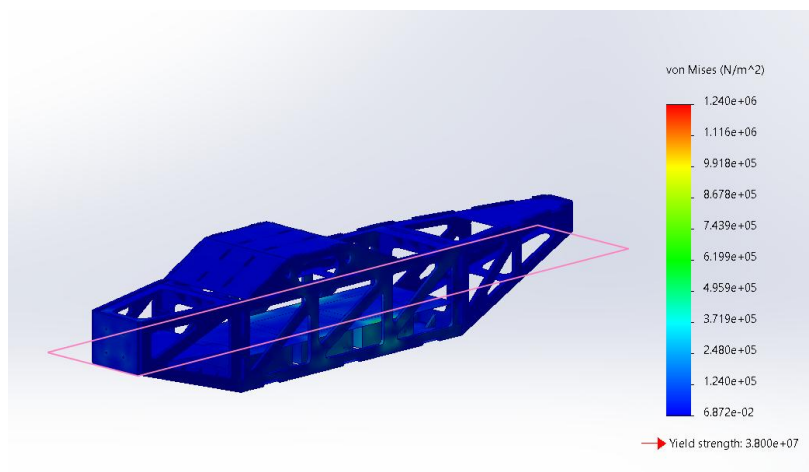


Figure 28 Stress on fuselage during hard landing

For both scenarios, the stress is significantly below the yield strength of the balsa wood, meaning that there is still a lot of room for weight reduction. However, due to time constraints and manufacturing difficulties, the iteration is not conducted. In the future, DBF team members can work on the CAD to optimize the weight and structure of the fuselage.

7.1.6 CFD of Different Fuselage (Prepared by Tsang Siu Tung)

We proposed two types of design, one with a bigger nose, another one with normal size. The CAD of the simplified design is shown in the figures below.

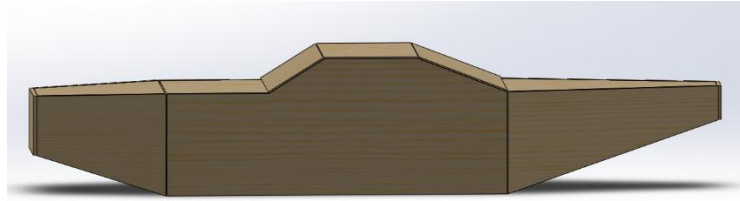


Figure 29 Simplified design of bigger nose fuselage

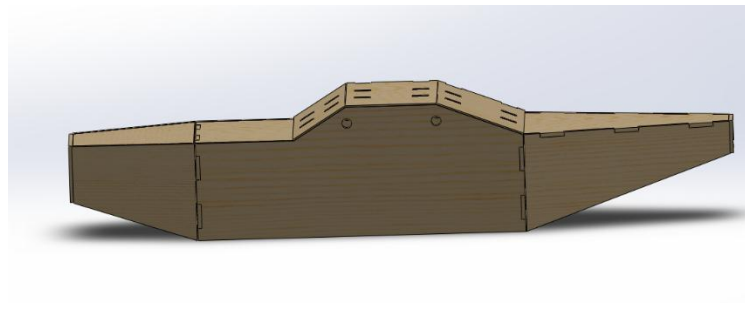


Figure 30 Simplified design of normal fuselage

To know which design has a better aerodynamic performance, a CFD is done for both designs. The flow velocity is set to 30 m/s. The k-omega SST model was chosen since it has a high near-wall accuracy. The drag force plot is shown below.

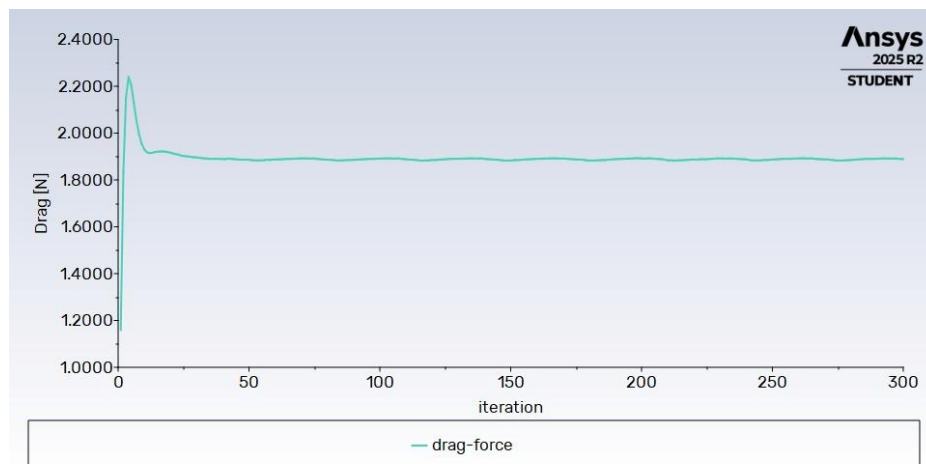


Figure 31 Drag force plot of bigger nose fuselage

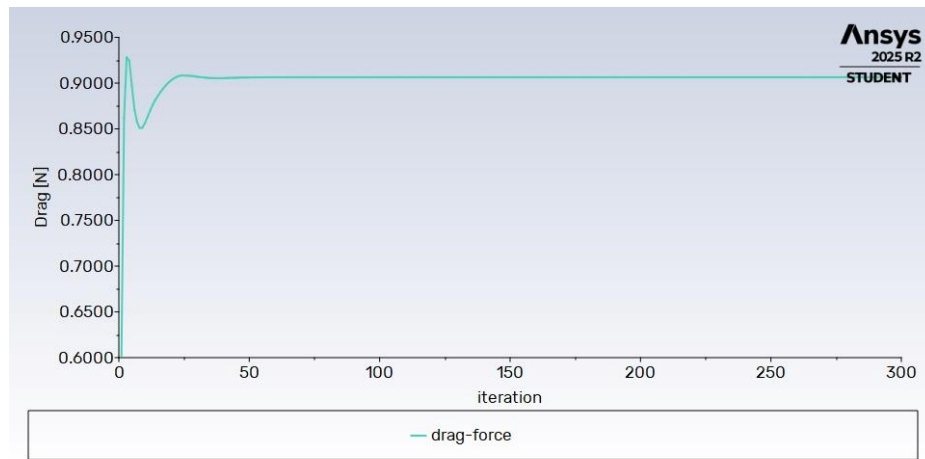


Figure 32 Drag force plot of normal sized fuselage

From the drag force plots, we can observe that shorter tail is a better design for aerodynamics. In fact, a larger nose leads to a significant increase in drag (more than 3 times), which is not worth using. Therefore, we keep the normal dimensions for our final design of fuselage.

7.1.7 Fuselage layout

The fuselage can be divided into 5 compartments, this section will discuss their design and features.

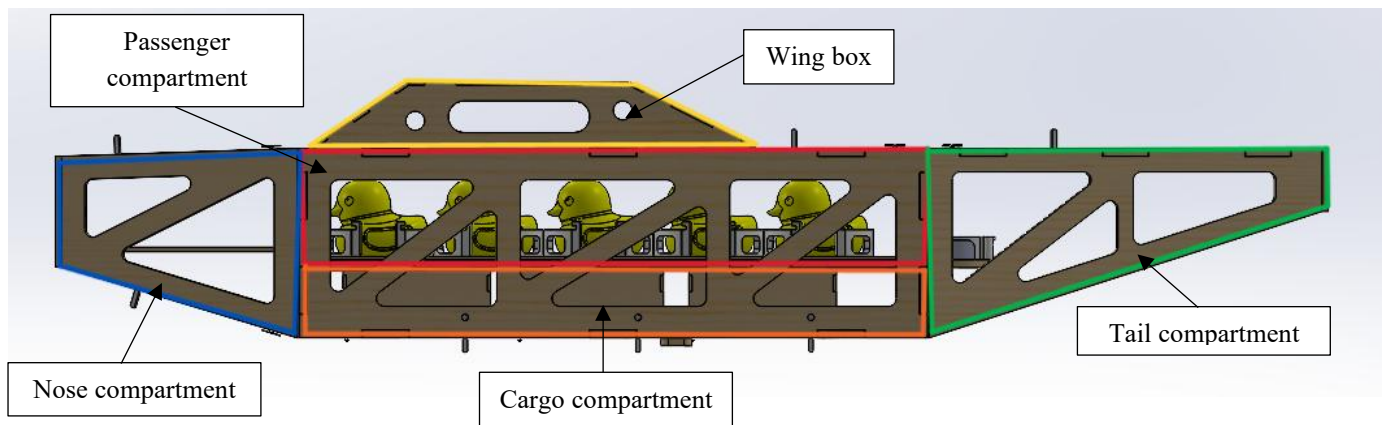


Figure 33 Fuselage layout

Passenger compartment

The passenger compartment is located at the centre part of the fuselage, it can hold 10 passengers on passenger seats. The passenger seat is a 3D printed holder to hold the passengers in place during flight. It has a rubber band on top to trap the passenger and allows easy insert and removal of passengers. It only weight 10 grams.

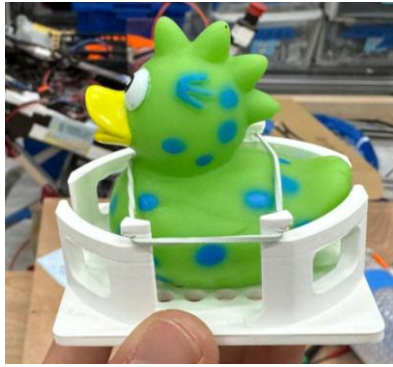


Figure 34 Passenger and seat



Figure 35 Rail inside passenger compartment

Within the passenger compartment, there are guide rails use to guide and secure the passenger seats. Additionally, a small pulley is installed to help the user retrieve passengers stored deep inside the fuselage. This device is pivotal as the DBF competition rule does not allow contestants to tilt the plane to slide the passengers out of the fuselage. The pulley is attached to a 3D print on the guide rail to pull the passenger seats.



Figure 36 Small pulley



Figure 37 3D printed pulling block

A trap door is installed towards the rear of the passenger compartment to access the compartment. The trap door make use of metal hinges and a pair of 3d printed quick release pin mechanism to make the trap door convenient to use and rigid. This door design is applied to all compartment doors on the fuselage



Figure 38 Quick release pin



Figure 39 Passenger compartment door

Cargo compartment

To retain structure rigidity and comply with the rule, 3 cargo compartments are designed at the bottom of the fuselage Acoording to the rule book, each cargo compartment can only have 1 door. However, the retrieval of cargo is challenging with only one door since a human hand cannot fit into the compartment. To improve the score in the ground mission, the cargo compartments were divided using 2mm wood plates, each cargo compartment now only holds one cargo. Wooden stoppers were designed on the compartment doors to hold the cargo in place.

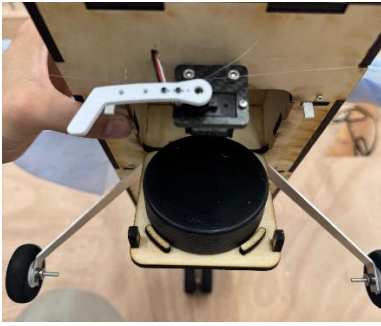


Figure 40 cargo



Figure 41 Cargo compartment door



Figure 42 Cargo compartments

Nose compartment

The nose compartment is split into 2 sub-compartments. The top one has larger volume is used to store the battery and fuse. The bottom one is used to store the ESC. The motor mounting plate not only have holes for mounting the motor but also have holes for wiring the motor.



Figure 43 Upper nose compartment



Figure 44 Lower nose compartment

Tail compartment

The tail compartment stores the pulley for pulling passengers and other avionics. A trap door allows easy access to this compartment.



Figure 45 Tail compartment door

Wingbox

The wingbox is located at the top of the fuselage, its main function is to provide rigid connection to the wing. The wing spars (Carbon fiber rods) are designed to be inserted into the wing box. There is a hole on each the carbon fibre rod, which aligns with a hole on a 3D printed PLA block. During wing installation, a screw can be inserted through the PLA block and the carbon fibre rod to lock the wing with the fuselage. A threaded metal insert is installed on the boom of the PLA block to hold the inserted screw in place. Thin balsa wood plates are installed at the top of the wing box to increase aerodynamic efficiency, The wingbox have slots to allow wiring between microcontroller and control surfaces on the main wing.

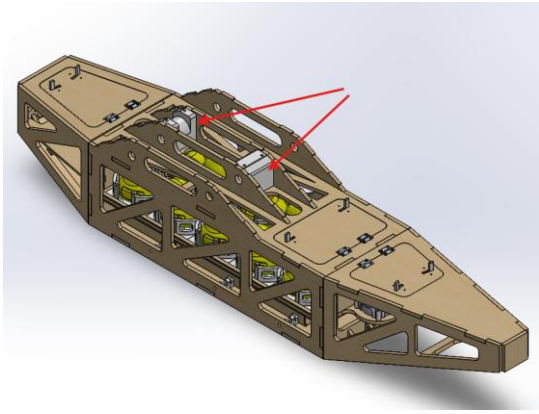


Figure 46 Wing locking 3D print

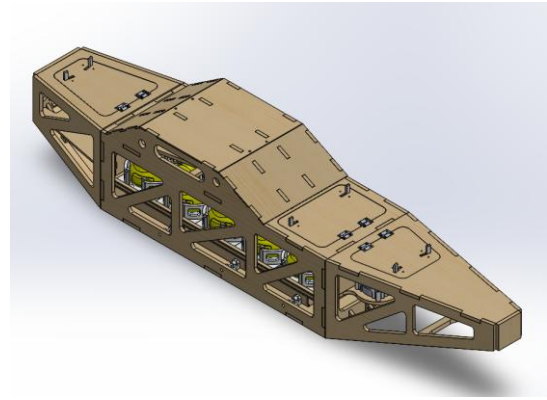


Figure 47 Closed wing box

7.1.8 Manufacture and assembly

The fuselage is manufactured using balsa wood in Innowing. For plates that need to be glued together, the contact areas are engraved to allow easy and accurate alignment. In addition, small components with similar shape are numbered such that the correct component can be identified during assembly.

For glue, AA glue is selected since it has strong tensile strength. According to the manufacturer, the glue can hold up to 2 MPa of tension.

7.2 **Wing (prepared by Gurung Srazeen)**

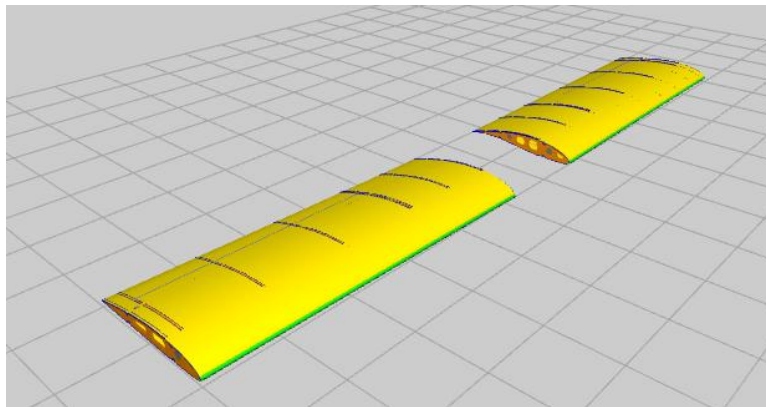


Figure 48 Model of the wing in Wing Helper

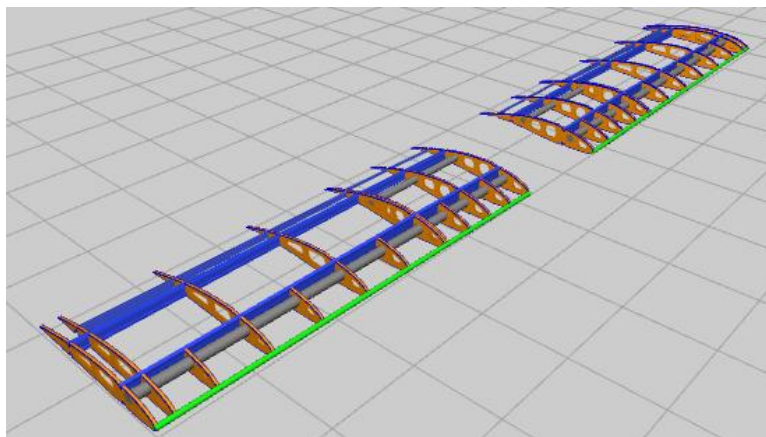


Figure 49 Model of the wing without sheeting

Building upon the previously defined wing geometry, a rectangular planform with a constant 200 mm chord and a 1200 mm span, the detailed structural and aerodynamic design was methodically segmented into two distinct functional regions to streamline both analysis and manufacturing. The inboard section, designated

the "Flap Region," is structurally optimized to accommodate the aerodynamic loads and mechanical integration of the high-lift flap system. Conversely, the outboard "Aileron Region" integrates the aileron control surface and culminates in the wingtip. The entire three-dimensional wing model was developed using the specialized aircraft design software WingHelper, which provided an efficient platform for lofting aerodynamic surfaces, defining internal rib/spar layouts, and generating manufacturing-ready drawings and cut files, significantly accelerating the transition from design to physical fabrication.

The Inboard Flap Region

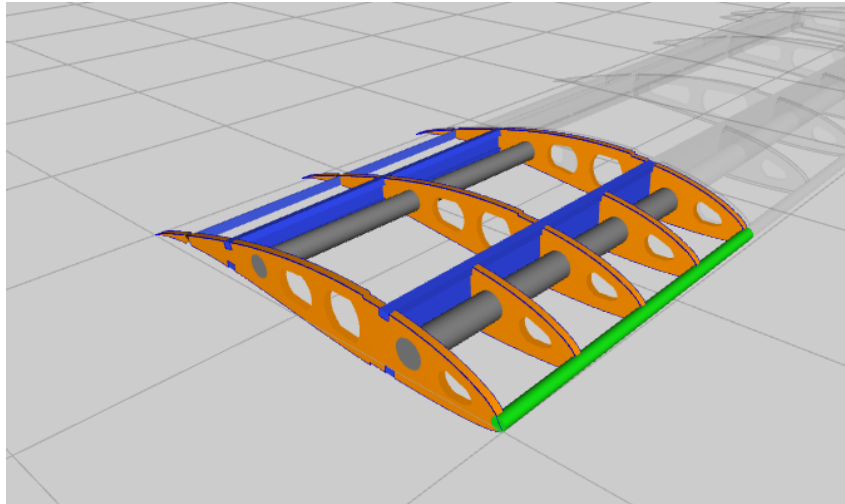


Figure 50 Inner model of the flap region

The inboard Flap Region, spanning a panel length of 160 mm, is constructed entirely using the Wortmann FX 63-137 airfoil, selected for its high lift characteristics at low Reynolds numbers. The primary structure consists of five 3 mm balsa wood ribs, two of which are strategically implemented as half-ribs extending 20% of the chord from the leading edge. This design achieves a dual purpose: significant weight reduction and the provision of a rigid, aerodynamically smooth contour for the covering material.



Figure 51 Drawing of the half rib

The primary load-bearing structure is a hybrid spar system designed for optimal load distribution. The main carbon fiber tubular spar, positioned at 20% chord, provides the primary bending stiffness, anchored near the aerodynamic center for efficient load transfer. A secondary carbon fiber tube at 50% chord works in concert to distribute bending loads and enhance torsional resistance. These tubular spars are integrated with a rectangular box spar and shear webs spanning from 25% to 75% chord, forming a robust I-beam configuration that encloses the central wing section. This geometry provides a massive increase in torsional rigidity by creating a closed-cell structure to manage shear forces effectively. A solid 3 mm balsa leading edge is incorporated to complete the aerodynamic profile, offering impact resistance and additional torsional stiffening while providing a firm substrate for the wing covering.

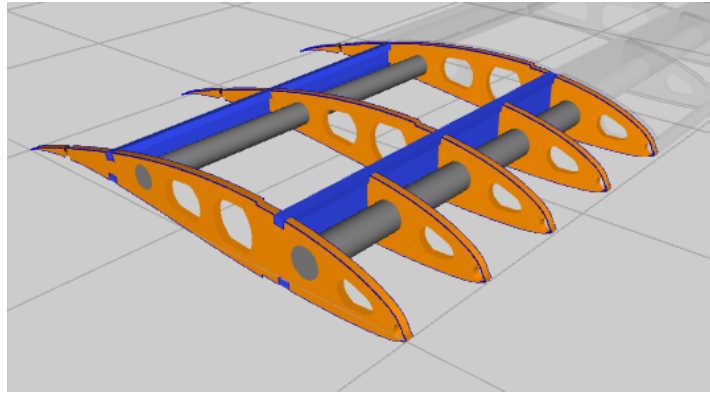


Figure 52 model of the ribs, CF rods and rectangular box spar

The slotted flap itself was designed using WingHelper's specialized "Aileron Cutter (K-form)" function, which generates a more aerodynamically effective shape than a simple hinged flap. Its custom hinge mechanism, designed in SolidWorks for optimal load bearing and minimal friction, will be fabricated via 3D printing to ensure precise geometry.

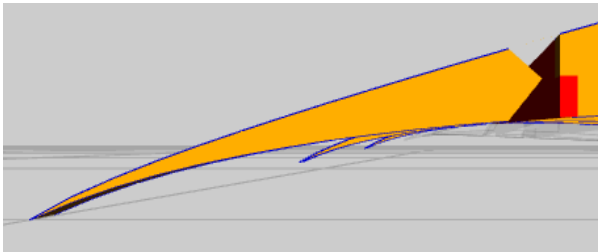


Figure 53 Flap using the "Aileron Cutter (K-form)" function

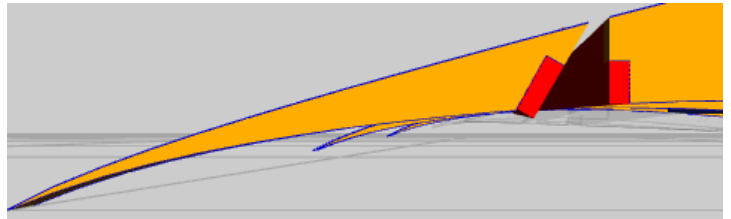


Figure 54 Flap using the "Aileron Cutter (simple flap)" function

Further weight optimization is achieved through lightening holes in the full ribs. To ensure sufficient torsional stiffness across the structure, 30% of the wing's surface, primarily around the leading edge and forward section, is skinned with 1 mm balsa sheeting. Finally, the entire assembly is sealed and finished with a heat-shrink plastic film, which creates a smooth, monolithic aerodynamic surface to promote laminar flow.



Figure 55 Drawing of the ribs showing holes for carbon fiber rod and lightening

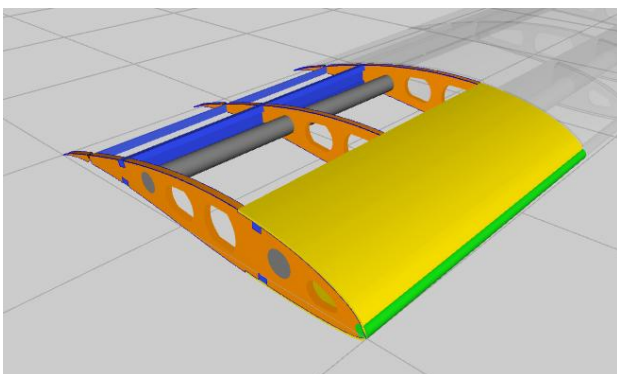


Figure 56 Wing model with 1mm balsa sheeting

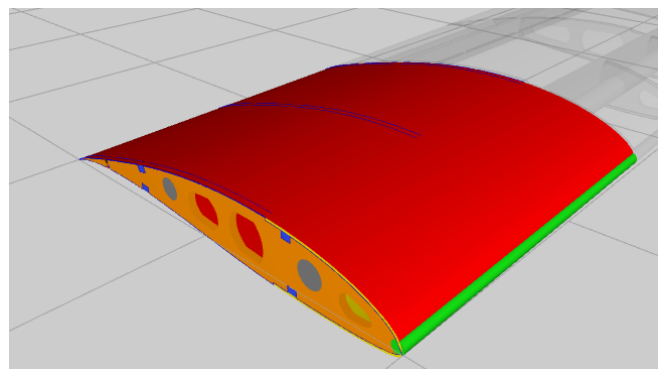


Figure 57 Wing model with heat shrinking plastic

The Outboard Aileron Region

The outboard Aileron Region, with a panel length of 365 mm, maintains structural and aerodynamic continuity from the inboard section. It is constructed with eight 3 mm balsa ribs, three of which are half-ribs extending from the leading edge for weight savings and contour definition. The load-bearing structure is a simplified yet robust adaptation of the flap region's design. It retains the primary carbon fiber tubular spar at the 20% chord location to ensure consistent bending stiffness and load transfer across the wingspan. However, to optimize weight in this region where torsional loads from control inputs are more dynamic, the secondary carbon fiber tube is omitted. Torsional rigidity is instead provided by the rectangular box spar and shear webs, which remain in place from 25% to 75% chord, creating the essential closed-cell structure.

The aileron control surface is integrated from the wing trailing edge using WingHelper's "Aileron Cutter (simple hinge)" function, extending to the seventh rib. This simple, lightweight hinge design is suitable for the aileron's required range of motion. The region is completed with 1 mm balsa sheeting over the forward 30% of the chord for torsional stiffening and a final heat-shrink plastic film covering to ensure a seamless, aerodynamic surface that extends to the wingtip.

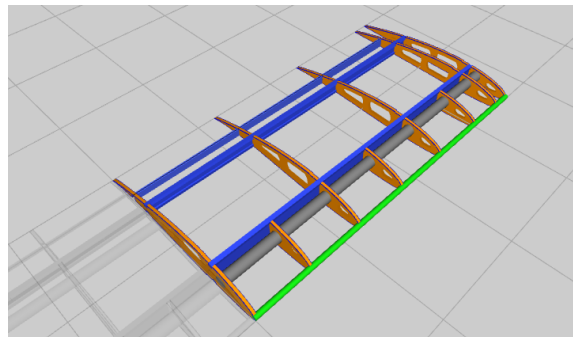


Figure 58 Model of the aileron region

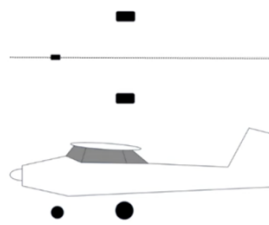


Figure 59 drawing of the first rib in aileron region

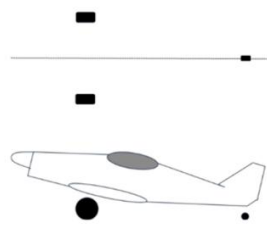
7.3 Landing Gear (Prepared by Lo Tsz Kin)

7.3.1 Landing gear configuration

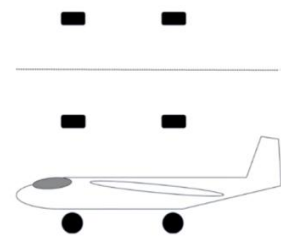
Landing gear configuration is an essential design phase for safe and stable ground handling, takeoff, and landing. It must support the aircraft's weight with full payload during all ground operations, provide stability during taxiing and steering, and absorb impact loads upon touchdown. Besides, it must provide enough height for the fuselage from the ground to provide sufficient space for cargo loading, which is determined as must more than 10 cm. We assess the relative advantages and disadvantages for tricycle gear, taildragger and quadricycle with respect to ground stability, structural complexity, weight, aerodynamic drag and integration with the overall aircraft layout.



TRICYCLE
Figure



TAILDRAGGER
Figure



QUADRICYCLE
Figure

7.3.2 Format comparison

We investigated the three conventional landing gear formats to understand their suitability with the overall aircraft design, especially the cargo loading and unloading mechanism and the banner deployment mechanism.

1. Tricycle format, with a nose gear located at the front of the fuselage and a main gear with usually two legs located at behind the center of gravity of the aircraft. Tricycle gear is a conventional and common design that can be applied for many flying conditions, such as usual airline aircraft. This design can provide more space under the fuselage to adapt to the door of cargo loading/unloading mechanism and the banner carrying and deployment mechanism. It also provides moderately high load support and relatively stable ground handling ability due to adaptability to steering mechanism. It has a few disadvantages which are it has higher structural weight and higher aerodynamic drag due to three highly load and moderately large legs are involved.
2. Taildragger format, with a main gear located at the front and a small taildragger near the end of the tail section. Taildragger format is also a conventional design that is usually applied on light aircraft. Due to the combination of small taildragger and a moderately high main gear at front, aircraft with taildragger design would has an inclined fuselage with the propeller pointing diagonally upward, which create a larger AoA for easier takeoff. Besides, this design is lighter and causing less aerodynamic drag because small and light taildragger, which indicates that the aircraft can have higher flying speed. However, taildragger format has few critical disadvantages that may significantly influence the flying performance. Due to a larger AoA, the aircraft would firstly experience difficult landing. Secondly, due to the short fuselage and long banner rod, when the aircraft was placed on the ground, it is easily predicted that the banner rod would touch the ground and further leads to unstable takeoff and chance of breaking the banner and the banner mechanism. Thirdly, taildragger is difficult to change direction of ground movement because steering mechanism cannot in installed to either fixed main gear or taildragger.
3. Quadricycle, with two main gears of total four legs in a rectangular configuration installed at the bottom of the fuselage, which one is located at front, and another is located at back. Quadricycle format is not as common as the other two format because it often used in large scale and heavy planes that required additional load support. Its most significant advantages is it can provide almost highest load support among all formats of landing gear due to four legs are involved. It also provides stable forward ground movement due to aligned rectangular wheels arrangement. Besides, it is relatively simple to purchase because only two standardized main gears are needed. However, quadricycle format has a few disadvantages. Firstly, it is the heaviest design among all, which may affect the flying performance. Secondly, it would generate highest aerodynamic drag due to four large scale legs. Thirdly, because the wheels need to be fixed at two main gears with a rectangular configuration, it is not feasible to install steering module to the aircraft, which leads to low ground maneuverability. Fourthly, quadricycle format would occupy most fuselage bottom surface area, which causes significant interruption with the cargo storing mechanism although it provides sufficient space for banner.

The decision matrix below would shortly indicate the relative advantages and disadvantages of the three formats with measurable grade and weighting. The characteristics with higher weighing means that it is more important factor. The higher grade is the more desirable factors, i.e. 5 is the best and 1 is the worst.

Characteristics	Weight	Tricycle	Taildragger	Quadricycle
Load support	4	4	3	5
Weight	3	4	5	3
Maneuverability	3	4	3	3
Propeller clearance	4	4	5	4
Cargo clearance	5	4	4	2
Banner clearance	5	5	2	5
Drag	4	3	4	3
Total score		113	102	101

Table Decision matrix of landing gear format

From the decision matrix, it is found that tricycle format is most suitable for current aircraft design and achieving missions' requirement.

7.3.3 Main gear format

There are two choices for constructing the main gear, which are fixed main gear and retractable main gear. The difference between them would be discussed in this section and the choice would be confirmed.

Fixed main gear

Fixed gear is a simple design that only involve a fixed component to connect the aircraft and the wheels. It is common to see fixed gear design on light aircraft. Its greatest advantage is this simple design can greatly decrease the difficulty to manufacture and install on the condense fuselage. However, because the gear would keep extended from the aircraft during the flight, it would increase the aerodynamic drag of the plane which would slightly decrease the flight speed.

Retractable main gear

Retractable gear is a complex design involving a gear retracting mechanism, usually motor and a gear storing mechanism. It is usually applied to airline aircraft. It requires a certain amount of space to store the gear in the fuselage or in wing. It also requires electronic system to control the retract and release of the gear. Its advantage is it can reduce aerodynamic drag due to hidden gear under aircraft surface. However, it has several shortcomings. It is complicated design involving extra system, so it is heavier, occupying more fuselage volume. Besides, it requires maintenance and have possibility of broken down.

The decision matrix below would shortly compare their features and further determine the desired design. The characteristics with higher weighing means that it is more important factor. The higher grade is the more desirable factors, i.e. 5 is the best and 1 is the worst.

Characteristics	Weight	Fixed	Retractable
Weight	3	4	2
Simplicity	3	5	2
Volume	4	4	2
Drag	2	2	4
Total score		47	28

From the decision matrix, fixed gear is a better choice considering condense fuselage.

7.3.4 Shape of main gear

There are two shapes for the main gear, which are curved and sharp cornered. Due to different structure of them, they have slightly difference performance in deformation subjected to forces.

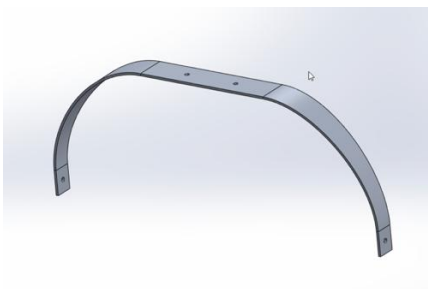


Figure Curved

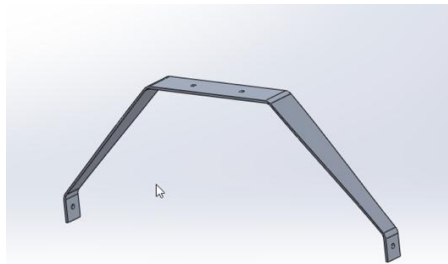


Figure Sharp cornered

After conducting a FEA in SolidWorks simulation with same material, aluminum 6061 alloy, it is found that under same magnitude of force (12N) exerted from the bottom of the legs, curved gear has slightly smaller degree of deformation than sharp cornered.

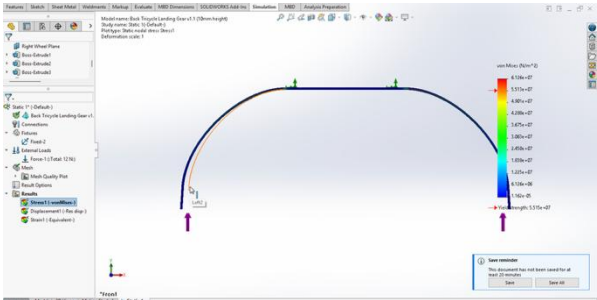


Figure Curved

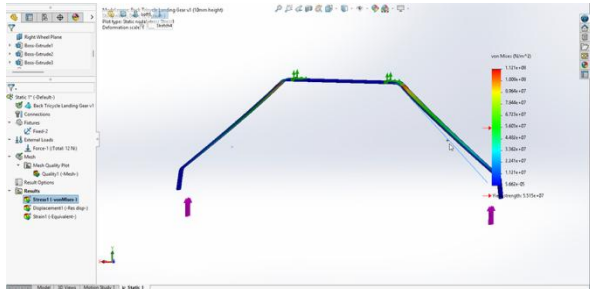


Figure Sharp cornered

In the simulation, it is found that most part of curved gear experiencing von Mises stress value that lower than the yield strength ($5.5515 \times 10^7 \text{ Nm}^{-2}$), while there is only half of the sharp cornered gear has von Mises stress value that lower than the yield strength. This result indicated that curved gear would have lower risk of material yielding and failure under shock than sharp cornered gear. Hence, curved gear is suggested for the final design to obtain higher shock absorbing ability.

However, it is found that most curved gear that we can purchase in Taobao store is made from carbon fiber with no holes drilled. The holes need to be drilled by hands, which may cause health issue due to small carbon particles, such as lung problem. Besides, curved gear is generally more expensive than sharp cornered gear by around twice. Hence, sharp cornered gear is suggested and chosen for first rapid prototype.

7.3.5 Material considerations

All landing gear in Taobao store is made of either aluminum 6061 or carbon fiber. Each offers distinct advantages and trade-offs that need to be evaluated against specific requirements of the aircraft and specific design phases.

Aluminum 6061 alloy

Aluminum alloys are prized for their well-understood properties, high ductility, and ease of manufacturing. They are readily machinable, weldable, and repairable with common workshop tools. Aluminum provides excellent strength-to-weight ratio among metals and predictable failure modes. However, its isotropic nature often leads to heavier structures compared to composites when stiffness-driven designs are required.

Carbon fiber

Carbon fiber offers a superior specific strength and specific stiffness, allowing for significant weight savings in carefully designed components. While carbon fiber can yield exceptionally lightweight and rigid structures, it requires more complicated manufacturing processes, which leads to higher cost and difficulty in rapid repairing. When carbon fiber is broke, the carbon particles produced would cause great health problem by touching skin and entering respiratory system.

The choice between these materials depends on the specific requirement of the landing gear in different design phase. For the very first prototyping, aluminum is suggested due to its lower cost and higher repairability while providing enough shock absorption ability. For the final aircraft which the overall design is confirmed, carbon fiber is suggested to provide higher strength and lower weight.

7.3.6 Nose gear configuration

Nose gear is an essential component in tricycle landing gear format because it absorbs shock force alone. There are two choices for nose gear, which are usual nose gear and spring nose gear.

Usual nose gear

Usual nose gear is very light that can greatly reduce the total weight of landing gear. However, its absorption degree depends on the stiffness and elasticity of the material, shape, and the mounting hole strength.



Figure 60 Usual nose gear

Spring nose gear

Spring nose gear conduct shock absorption through compression or extension of spring. Hence, the aircraft can land with less vibration and sudden deviation in movement due to the converting from shock energy to elastic energy stored in spring. However, it is complicated that required assembly for pillars and spring.

The decision matrix below would shortly indicate the relative advantages and disadvantages of them. The characteristics with higher weighing means that it is more important factor. The higher grade is the more desirable factors, i.e. 5 is the best and 1 is the worst.

Characteristics	Weight	Usual gear	Spring gear
Stability	4	2	4
Weight	2	4	3
Simplicity	2	4	3
Total score		24	28

Table 27 Decision matrix of types of landing gear

From here, it is determined that spring nose gear is a desirable choice to provide more stable landing.

Spring gear can be divided into two types, which are compression spring and extension spring.



Figure 61 Compression spring gear



Figure 62 Extension spring gear

Compression spring gear

In a compression-type landing gear, the primary shock-absorbing element is loaded in compression during landing. As the wheel strikes the ground, the strut or spring shortens like a pogo stick or an automotive suspension coil. This design is frequently used for it handles axial loads and steering forces well. It typically uses springs that compress under load.

Extension spring gear

In an extension-type landing gear, the main structural member is loaded in tension and bending upon impact. The gear leg, often a leaf spring, flexes or stretches to absorb energy. This design is frequently used for main gear on taildragger aircraft and many lightweight UAVs because it is simple, lightweight, and easy to fair aerodynamically.

The decision matrix below would shortly indicate the relative advantages and disadvantages of them. The characteristics with higher weighing means that it is more important factor. The higher grade is the more desirable factors, i.e. 5 is the best and 1 is the worst.

Characteristics	Weight	Compression spring	Extension spring
Stability	4	4	3
Simplicity	3	3	4
Cost	1	3	4
Total score		28	28

From the decision matrix, it is found that the two design has generally same grade point, which compression spring gear is better in landing and extension spring gear has lower cost for about half of compression one. The current design chose using extension spring gear to reduce the cost and provide sufficient landing shock-aborting ability to collect real life flight data. Once the data is collected, extension spring can be changed to compression spring gear to achieve more stable landing experience.

The current extension spring gear can provide approximately 1900g (18.6N) force absorption.



Figure 63 Nose gear force absorption

It should be mentioned that spring nose gear and main gear in Taobao store although same size number is written, their actual vertical size is different, which nose gear is higher than main gear. But due to the inclined fuselage nose, the fuselage would still be roughly horizontal.

7.3.7 Steering mechanism

Steering mechanism describes the directional control of an aircraft during ground operations, including taxiing, takeoff roll, and landing rollout. It is typically achieved through nose-wheel steering linked to the servo motor. Effective steering is essential for safe runway alignment and maneuvering in confined areas.

The steering mechanism was purchased on Taobao. The kit includes a CNC mounting part to mount the nose gear and servo motor on the fuselage. A 9g A0090 servo motor provided by Dr. Chan is suitable for this kit.



Figure 64 Steering module

However, the current fuselage head design is relatively small and dense, which cannot provide sufficient space for installation of steering module. Hence, steering module is decided not to be installed on the current design, or a new small steering module should be introduced.

7.4 **Banner Towing (Prepared by Tsang Siu Tung)**

7.4.1 Banner Dimensions

According to Rulebook 3.1.3, the banner must have a minimum length of 10 inches and a maximum aspect ratio (length/height) of 5. For prototyping, we will take the rectangular banner width as 1m and length as 5m.

7.4.2 Banner Drag Calculation

By using the equation below, we can determinate the drag force exerted by the banner.

$$F_D = \frac{1}{2} C \rho_f A v^2$$

where C is the drag coefficient,

ρ_f is the density of the fluid (air in this case),

v is the velocity of the object,

A is the projected cross-sectional area of the object in a direction perpendicular to the object's motion.

Assume the banner has negligible thickness. Take $C = 0.0364$ for a rectangular banner and airspeed as 29m/s. The density of air is 1.225 kg/m³. We have

$$F_D = \frac{1}{2}(0.0364)(1.225)(5)(29)^2 = 93.8 \text{ N}$$

7.4.3 Banner Material Selection

Three types of cloth were selected for, including cotton cloth, nylon cloth and polyester cloth. Cotton cloth is commonly used in clothing, making it easy to obtain. Nylon cloth is used to make sportswear, known for its lightness. Polyester is used to make flags, which may suit our purpose of making a banner. To ensure a fair comparison, all three cloths for weighing are 1.5m x 2m. After weighing, we obtain the weights of each material.

Material	Weight (g)
Cotton	360
Nylon	100
Polyester	340

Table 28 Weights of three banner materials

From the table above, nylon cloth has the lightest weight. Therefore, we used nylon cloth as our banner material.

7.4.4 Banner Logo Printing

There are two common methods of imprinting HKU logos to the cloth – digital textile printing and hot stamping. For digital printing, a Chinese manufacturer used a printing machine to directly print HKU logos on the polyester cloth. Below shows the results of the banner.



Figure 65 Front of the polyester banner



Figure 66 Back of the polyester banner

For hot stamping, we purchased hot stamping stickers and iron them to the cloths by ourselves. The procedure of hot stamping is shown below. A household electric iron is used for this process.

1. Place the banner cloth on a hard and smooth surface.
2. Iron the banner cloth to ensure a flat surface for the sticker to adhere.
3. Place the sticker on the cloth and cover it with a cotton cloth.
4. Adjust the temperature of the electric iron (high for cotton, nylon halved the temperature of cotton)
5. Press the electric iron on the cloth firmly for 30 seconds and gradually move it to other parts of the sticker.
6. After ironing, slowly peel off the plastic sheet attached on the sticker.
7. In case there are any parts that do not adhere to the cloth, iron that part again.



Figure 67 Finished hot stamping on nylon cloth (front)

Figure 68 Hot stamping on nylon cloth (back)

If the temperature of the electric iron is set too high for nylon, some parts of the cloth will melt. On the other hand, if the temperature is too low for cotton, the sticker will not adhere to the cloth. As shown in the figures above, the hot stamping sticker adhered firmly to the nylon cloth. It cannot be peeled off easily. However, it is recommended to prepare multiple banners for actual competition since nylon cloth is very thin to be torn apart by accident.

For logo printing methods, there are two main considerations –requirement of the rules and weight. According to Rulebook 3.1.3: The banner must have the team’s university name or logo on both sides. From digital textile printing, the HKU logos on the back are reversed. It is not clear that if the competition permits this. However, if we use hot stamping, we can imprint logos at both the front and the back of the cloth, enhancing the flexibility of design and manufacturing. Therefore, hot stamping is a better method for banner design.

Regarding the weight, the figure below shows that nylon cloth with the logo is 130g. Since it is very light, we can have the flexibility of either hot stamping another logo at the back or directly combining two banners into one.



Figure 69 Weight of the nylon cloth with hot stamped sticker imprinted

As a result, nylon cloth with hot stamping is chosen for banner. This combination is the lightest and fits the rules without causing arguments.

7.4.5 Banner Release Mechanisms

The configurations considered for banner release are direct release and slow release. Direct release consists of a locking mechanism that prevents the banner from deploying during take-off, whereas the slow-release mechanism consists of a 360-degree continuous servo motor and a 3D printed roller to gradually release the banner. We have constructed a decision matrix to decide which release mechanism is more favourable to winning.

Characteristics (All weighting = 1)	Direct Release	Slow Release
Stability	1	2
Time for release	2	1
Weight	2	1
Size (Volume occupied)	2	1
Score	7	5

Table 29 Banner Release Mechanism Decision Matrix (Highest: 2; Lowest: 1)

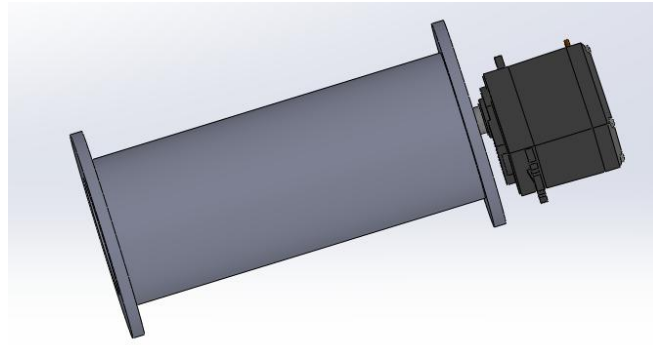


Figure 70 Design of slow release system

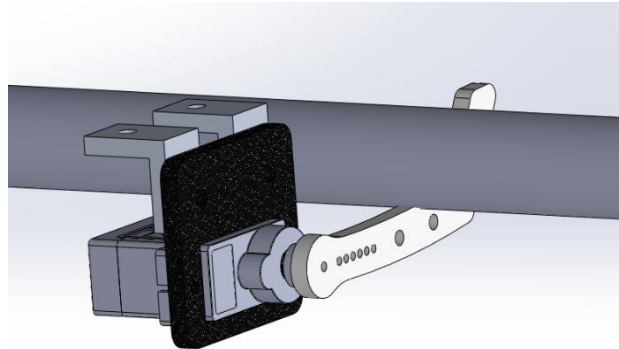


Figure 71 Design of direct release system

Hence, we will choose direct release system for banner release mechanism. Below shows a detailed description of the system design.

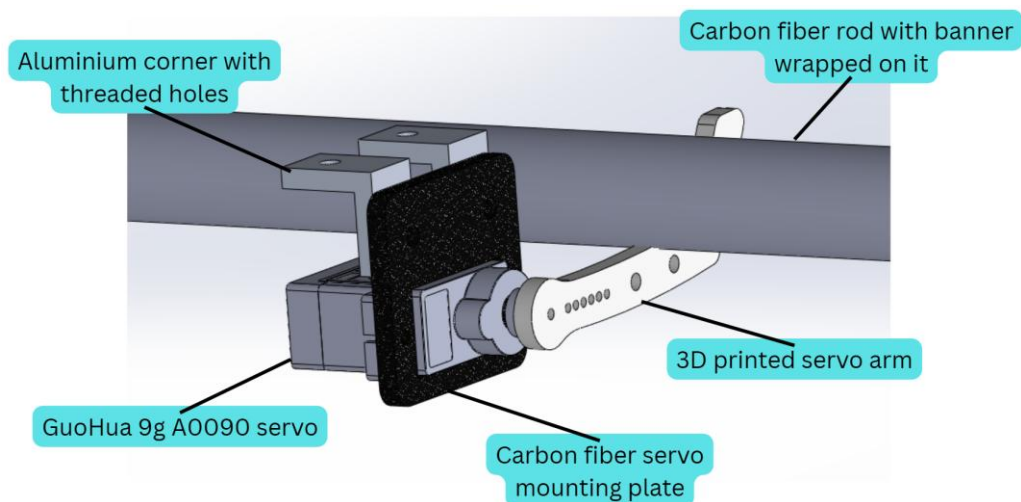


Figure 72 Direct banner release system design

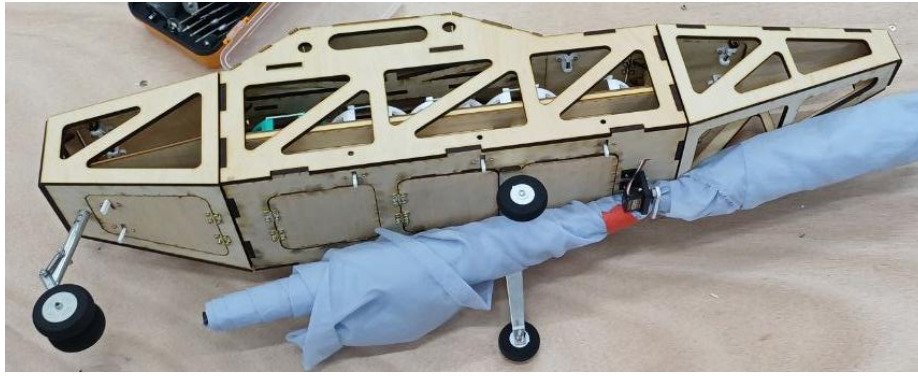


Figure 73 Banner release mechanism mounted on fuselage

The aluminium corners will be mounted to the fuselage. The threaded holes on the corners, the carbon fibre rod and the carbon fibre servo plate further minimized the weight of the system. To ensure structure rigidity, the carbon fibre plate used is 5mm thick, and the 3D printed servo arm is made of PLA+. Fishing wires connect the carbon rod of the banner to both the servo arm and the fuselage, such that the servo arm is only actuating the release motion rather than taking the drag force. The method to achieve this is simple – ensuring the length of fishing wires connecting to the servo arm is longer than that of fuselage.

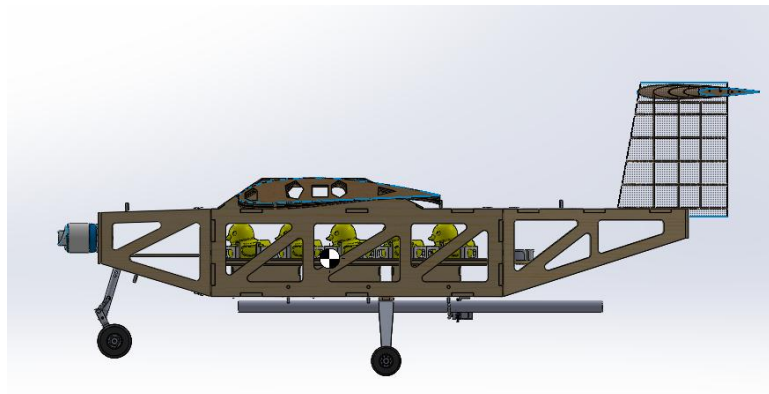


Figure 74 Whole plane with c.g. position

The position of the banner release mechanism can be moved forward to adjust the c.g. position. As long as the banner has sufficient clearance with the nose gear, the position of the mounting can be changed.

7.5 Empennage (by Perry)

7.5.1 Tail Configuration

The tail configuration of an aircraft plays a critical role in providing longitudinal and directional stability, as well as adequate control authority for pitch and yaw. Selecting an appropriate tail arrangement is therefore a key step in the conceptual design phase. We investigate different tail configuration where we consider the use conventional tail, v-tail and t tail. The objective is to assess their relative advantages and disadvantages with respect to stability and control, structural complexity, aerodynamic efficiency, weight, and integration with the overall aircraft layout.

1. The conventional tail, comprising a horizontal stabilizer and a vertical fin arranged in a cruciform layout, supported directly by the fuselage structure. It is well-understood, relatively simple to design and manufacture, and generally offers predictable stability and control characteristics, so it serves as

the baseline. Its primary disadvantages include not being provide the clean flow at cruise, making it less efficient comparatively.

2. The V-tail combines the functions of the horizontal and vertical stabilizers into two inclined surfaces. This configuration can reduce the total wetted area and, potentially, overall drag and weight. However, it introduces control coupling between pitch and yaw, requiring more complex control linkages and sometimes sophisticated mixing mechanisms. The need to accurately build 2 inclined surfaces also add significant complexity the structural design. This complexity makes it an undesirable option for this project.
3. In the T-tail arrangement, the horizontal stabilizer is mounted at the top of the vertical fin. This configuration places the tailplane outside the main wing wake for many flight conditions, potentially improving elevator effectiveness and reducing interference effects from the fuselage. On the other hand, the T-tail typically demands a stronger, heavier fin structure, and it may experience deep stall when the wing wake blankets the tailplane at high angles of attack. The improvement at cruising outweighs the increase in complexity of the structure, which is why we settle on this configuration.

7.5.2 Structural Material Comparison

The choice of structural material has a decisive influence on the internal design of the tail and must therefore be addressed immediately after selecting the tail configuration. Foam, balsa wood, and 3D-printed plastics has been chosen but each require fundamentally different internal arrangements, joining methods, and reinforcement strategies to achieve the necessary stiffness and strength. For example, a balsa tail typically relies on built-up ribs, spars, and shear webs, whereas a foam tail may use a solid or semi-monocoque core with embedded carbon spars, and a 3D-printed tail might employ thin shells with integrated internal lattice structures. These differences directly affect the design process, including the level of detail in the CAD model, the choice of manufacturing tools, assembly techniques, and the complexity of joints and connections. Most critically, material selection drives the weight distribution and overall mass of the tail, which in turn influences stability, control effectiveness, and the placement of the aircraft's center of gravity.

Criterion	Weighting (Max)	Foam	3D print	Balsa Wood
Weight	9	7	3	9
Strength-to-weight	4	2	2	4
Crash resistance	2	2	0	1
Ease of fabrication	3	3	2	1
Cost	1	1	0	0
Repairability	2	0	2	1
Availability	4	0	4	3
CAD Complexity	2	2	1	0
Total Score	27	17	14	19

Table 30 Decision Table for Tail Material

1. Balsa wood is a very light material, around 0.3g/cm^3 , with an excellent stiffness-to-weight ratio along the grain, making it strong when used in traditional built-up structures such as ribs, spars, and trusses. It requires conventional model-building skills—cutting, pinning, jigging, and gluing—and is more labour-intensive than foam or prefabricated 3D-printed parts. Repairability is generally good: damaged sections can be cut out, replaced with new balsa, and then recovered with film or tissue. Material cost is moderate, and when properly covered, balsa provides very smooth, aerodynamically efficient surfaces. Its main advantages are high performance, low weight, and the ability to tailor

strength locally by thickening spars. However, balsa structures are more time-consuming to build, less crash-resistant than EPP foam and often involve relatively complex joints and connections. As shown in the table above, balsa wood scored the highest among the materials which is why we chose it as the structural material of the empennage.

2. 3D-printed plastic as a structural material offers repeatable parts with very high design freedom, allowing internal ribs, spars, ducts, and hinges to be integrated into a single component—especially useful for complex geometries. It provides good local stiffness but is strongly anisotropic, being weaker between layers. However, it tends to be heavy of 1.2g/cm^3 , so poor design can easily result in an overweight model and is impossible to achieve strength-to-weight ratio of balsa wood. Innowing provide good printers making it very accessible. Damaged parts can be replaced individually and are repairable using glue or by welding with a soldering iron and filament. Films must be used to achieve a smooth aerodynamic surface.
3. Foam is a light material with low stiffness that bends easily, offering only moderate impact resistance and tending to creep or dent under load, which makes it less suitable for high-precision or high-performance airframes. It is easy to shape by hand through CNC, hot wire and sanding, though innowing do not provide CNC machines making it less of a viable material. While surface finish is naturally rough and porous, it can be improved with filler, tape, or film covering to achieve an acceptable aerodynamic surface. Foam is generally cheap and very forgiving in crashes, absorbing impacts without catastrophic breakage, and it is compatible with simple reinforcements such as carbon spars making it great for prototyping. Repair is straightforward using foam-safe CA, hot glue, UHU, epoxy, or tape, but when the structure is badly crushed or deformed, you often need to replace the whole part rather than perform fine structural repairs.

7.5.3 Sizing

The tables below give a reference to the form of the empennage. The horizontal tail volume and vertical tail volume dedicate the surface area of the tail and the aspect ratio and taper ratio determines the shape of the tail. We took $V_{HT}=0.5$ and $V_{VT}=0.04$ since it is a homebuilt plane and $A_H=4$, $\lambda_H=0.5$, $A_V=1$ and $\lambda_V=0.8$ for it is a T-Tail.

Table 9.1 Conventional aspect ratios A and taper ratios λ from empennages on transport category airplanes (**Raymer 1989**)

Type	Horizontal Tailplane		Vertical Tailplane	
	A	λ	A	λ
Conventional Tail	3.00 ... 5.00	0.3 ... 0.6	1.3 ... 2.0	0.3 ... 0.6
T-Tail	as Conventional Tail	as Conventional Tail	0.7 ... 1.2	0.6 ... 1.0

Table 30 31 Reference for Aspect Ratio and Taper Ratio

	V_{HT}	V_{VT}
Sailplanes	0.50	0.02
Homebuilt	0.50	0.04
General aviation - single engine	0.70	0.04
General aviation - twin engine	0.80	0.07
Agricultural aircraft	0.50	0.04
Twin turboprop	0.90	0.08
Flying boat	0.70	0.06
Jet trainer	0.70	0.06
Jet fighter	0.40	0.07
Military cargo/bomber	1.00	0.08
Jet transport	1.00	0.09

Table 32 Reference for Horizontal Tail Volume and Vertical Tail Volume

We then calculate the optimum tail arm with the equation below

$$l_T = \sqrt{\frac{2 \cdot S(V_{HT} \cdot c_{MGC} + V_{VT} \cdot b)}{\pi(R_1 + R_2)}}$$

Where R_1 = Radius of fuselage at tail wing
 R_2 =Radius of fuselage at main wing
 c_{MGC} = Mean Geometric Chord of main wing
 b = base of main wing
 S = Surface area of main wing

We found the optimum tail arm to be 0.317m which is smaller than the actual length between the main wing and the end of the plane. This allows us to have a even smaller tail if we can utilise the full 0.49m available as the tail arm, further reduce weight and drag induced.

Below is a table for the geometric calculation and value obtained that is related to the empennage

Parameter	Equation	Value	Unit
Horizontal Surface Area (S_H)	$S_H = \frac{V_{HT} \cdot S \cdot c_{MGC}}{l_T}$	0.0463	m ²
Vertical Surface Area (S_V)	$S_V = \frac{V_{VT} \cdot S \cdot b}{l_T}$	0.0222	m ²
Horizontal Tail Span (b_H)	$b_H = \sqrt{A_H \cdot S_H}$	0.373	m
Vertical Tail Span (b_V)	$b_V = \sqrt{A_V \cdot S_V}$	0.163	m
Horizontal Root Chord (c_{HR})	$c_{HR} = \frac{2b}{(1 + \lambda_H) \cdot A_H}$	0.166	m
Horizontal Tip Chord (c_{HT})	$c_{HT} = c_{HR} \cdot \lambda_H$	0.0829	m
Vertical Root Chord (c_{VR})	$c_{VR} = \frac{2b}{(1 + \lambda_V) \cdot A_V}$	0.151	m
Vertical Tip Chord (c_{VT})	$c_{VT} = c_{VR} \cdot \lambda_V$	0.121	m
Elevator Surface Area (S_{ele})	$S_{ele} \approx 0.3 \cdot S_H$	0.0139	m

Chord of Elevator (c_{ele})	$c_{ele} = \frac{S_{ele}}{b_H}$	0.0373	m
Rudder Surface Area (S_{rud})	$S_{rud} \approx 0.3 \cdot S_V$	0.00666	m
Chord of Rudder (c_{rud})	$c_{rud} = \frac{S_{rud}}{b_V}$	0.0409	m

Table 33 Calculation for Sizing of Tail

7.5.4 Geometric Checking

After completing the calculations, it is necessary to check whether the design is geometrically feasible. The chord of the elevator must be larger than the tip chord of the horizontal tail, and the chord of the rudder must be larger than the tip chord of the vertical tail. In addition, the elevator must not be obstructed by the vertical tail or rudder throughout its full deflection range.

$$c_{ele} < c_{HT}$$

$$c_{rud} < c_{VT}$$

$$c_{HT} - c_{ele} > c_{VT}$$

Then we check whether the location of the horizontal tail will result in a pitch-up at stall. The non-dimensional vertical distance is obtained by dividing the vertical tail span b_V by the mean geometric chord of the main wing c_{MGC} , giving a value of 0.9, and the non-dimensional horizontal distance is obtained by dividing the tail arm l_T by the mean geometric chord c_{MGC} , giving a value of 2.0. As shown in the graph below, this placement of the horizontal tail falls in the region where the aircraft experiences a pitch-up at stall with warning and lies outside the zone corresponding to pitch-up without warning, so the configuration is considered acceptable.

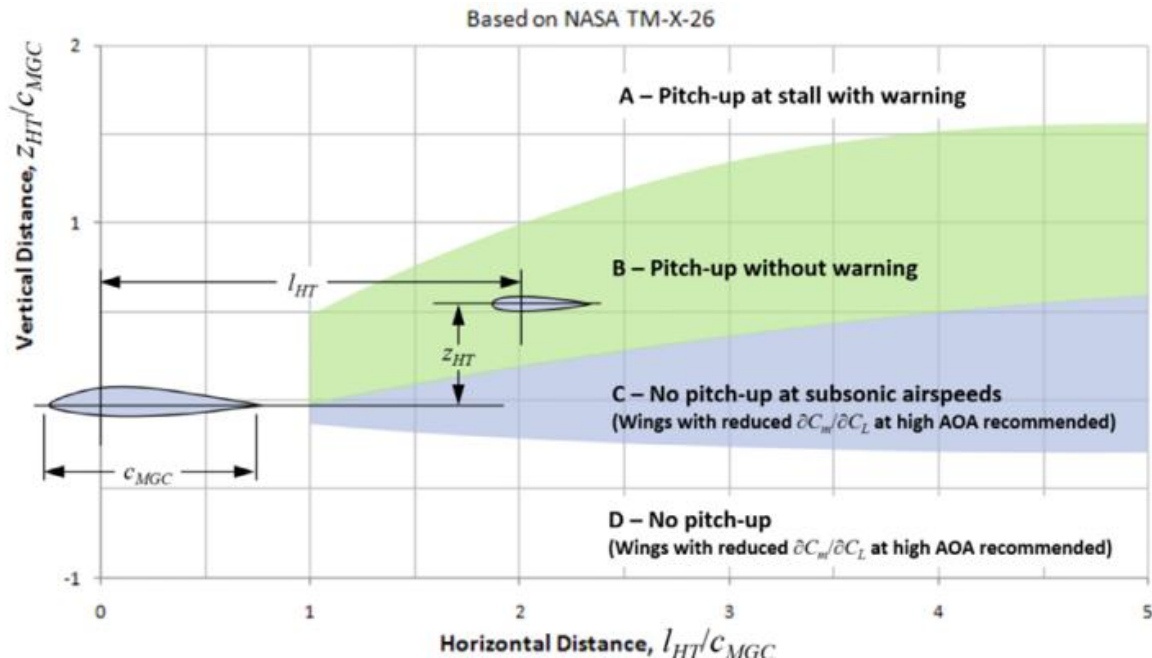


Table 34 Recommended placement of the horizontal tail behind the wing.

7.5.5 Aerofoil

An aerofoil is used for the tail instead of a flat plate because it provides more predictable, efficient, and controllable aerodynamic behaviour. An aerofoil has a smoother pressure distribution and a higher lift-to-drag ratio, so it can generate the required tail forces) with less drag and often with a smaller surface

area. It also has a more linear and well-defined lift curve and a gentler, more controllable stall, which is important for maintaining stable pitch and yaw behaviour near stall. In contrast, a flat plate produces more drag, stalls earlier and more abruptly, and gives less precise control forces, which can reduce stability and make trimming and handling less reliable, especially at higher angles of attack or higher speeds.

Here are three aerofoils we are going to compare: NACA 0012 and NACA 0008, which are recommended in the reference, and HT05, which is the thinnest symmetrical aerofoil we found. We compared their lift coefficient (C_L), drag coefficient (C_D), and angle of attack AoA characteristics. NACA 0012 shows a higher C_L at higher AoA, meaning it has a greater tendency to generate restoring moments on the aircraft. Its higher stall angle and higher peak C_L are also important advantages, especially for a T-tail configuration, which is more prone to entering a deep stall.

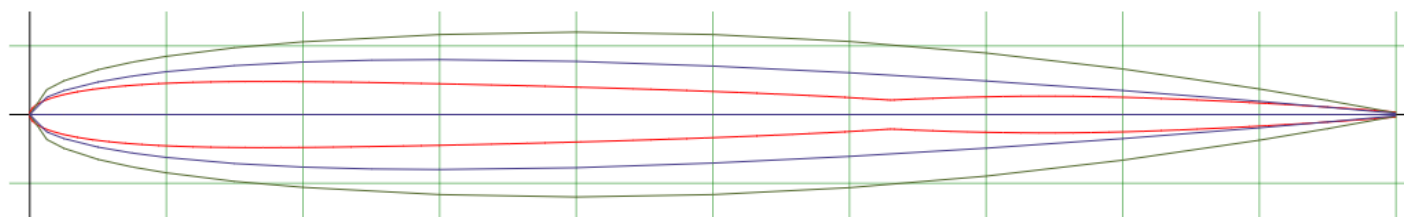


Figure 75 Profile of the 3 aerofoils

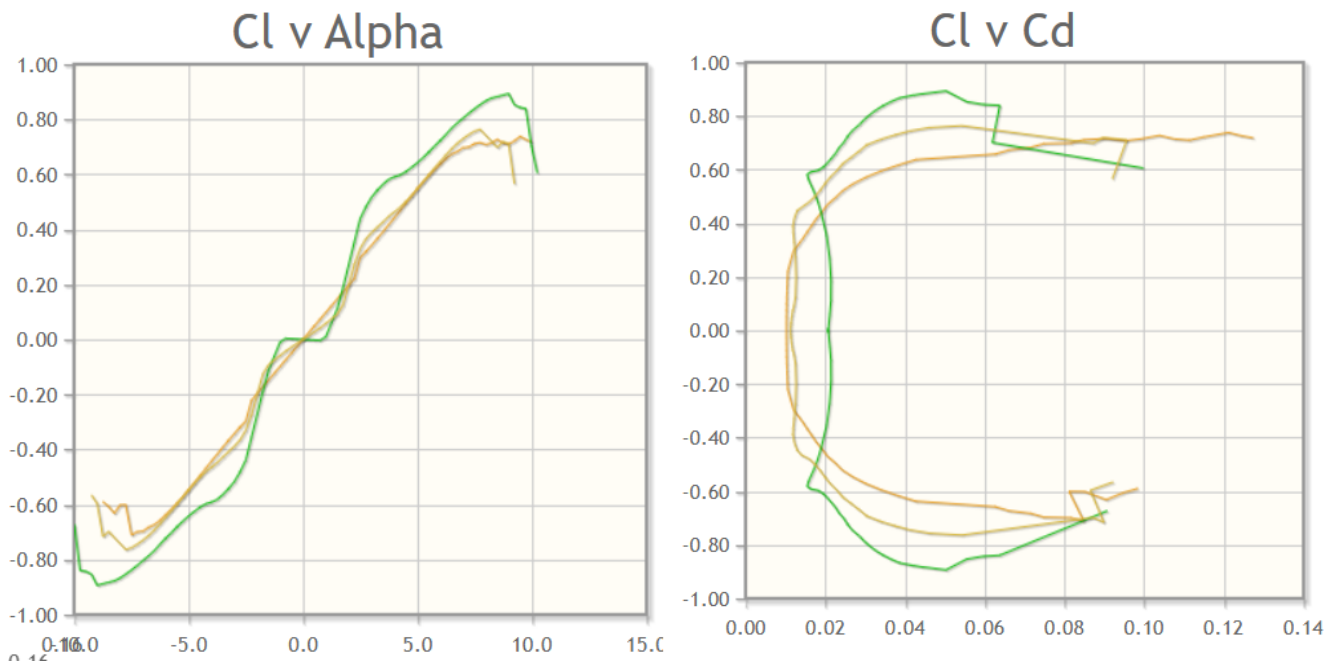
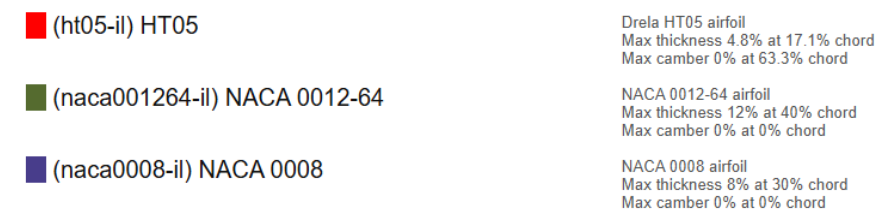


Figure 76 C_L v AoA between 3 aerofoils

Figure 77 C_L v C_D between 3 aerofoils

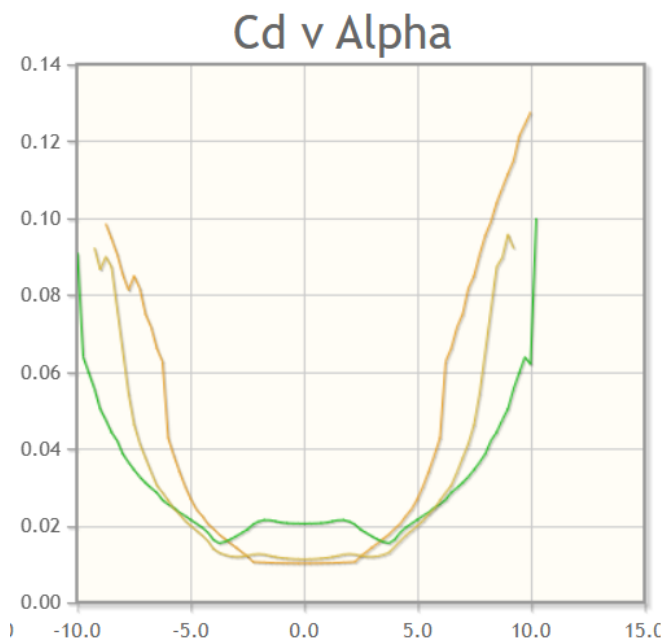


Figure 78 C_D v AoA between 3 aerofoil

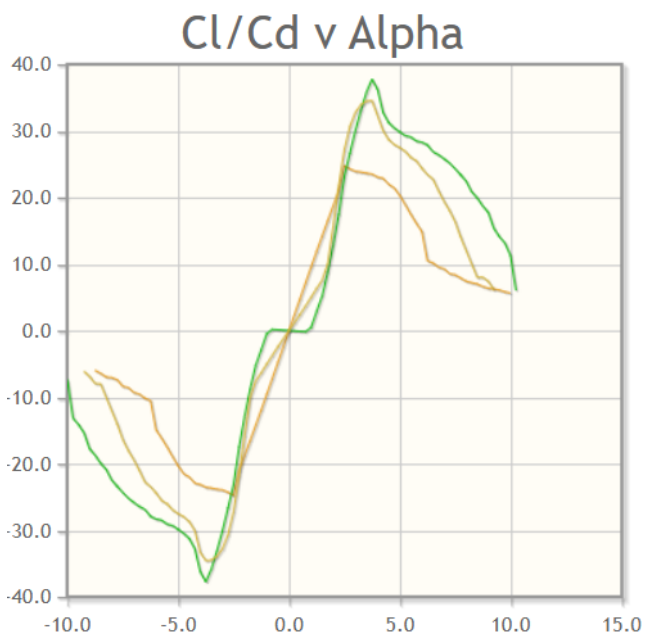


Figure 79 C_L/C_D v AoA between 3 aerofoils

☒	ht05-il	100,000	9	24.8 at $\alpha=2.5^\circ$	Mach=0 Ncrit=9
☒	naca001264-il	100,000	9	37.8 at $\alpha=3.75^\circ$	Mach=0 Ncrit=9
☒	naca0008-il	100,000	9	34.6 at $\alpha=3.75^\circ$	Mach=0 Ncrit=9

7.5.6 CAD

The empennage was designed to be joined using a cross-lap joint, which greatly increases the structural integrity of the tail, this method also stop the structure from sliding around and improves accuracy and speed during assembly. The structure was originally made from 1 mm balsa wood. However, several issues arose during manufacturing, leading us to switch to 2 mm wood. First, the pieces curled up during laser cutting, making them unable to withstand compressive and tensile loads. Second, the joint areas were too small to hold the parts securely in a fixed position. Finally, the material was too flexible, making it difficult to align the components to the correct shape.

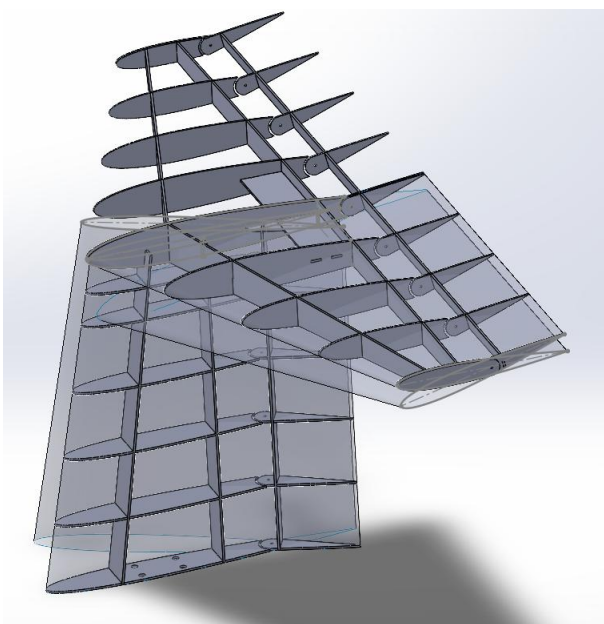


Figure 80 First prototype

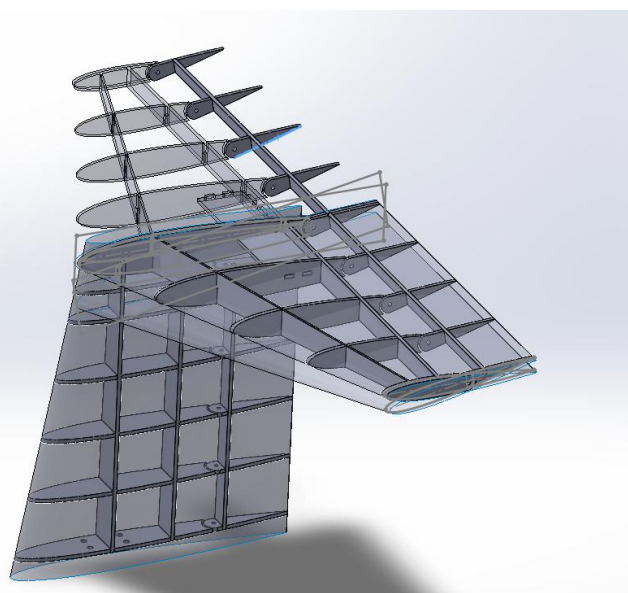


Figure 81 Final design

A platform is added to the horizontal tail via a tab-and-slot connection to provide a mounting base for the servo that controls the elevator. After switching to 2 mm balsa, all joints were redesigned to be at right angles, and the vertical tail structure was modified accordingly, which also has the added benefit of aligning the primary loads parallel to the supports, improving load transfer and stiffness. However, the horizontal tail could not be fully redesigned in the same way because diagonal members are required to leave sufficient space for the servo and its linkage. As a result, one of the diagonal supports must remain in 1 mm balsa to fit within the available geometry, so this area is identified as a structurally weaker.

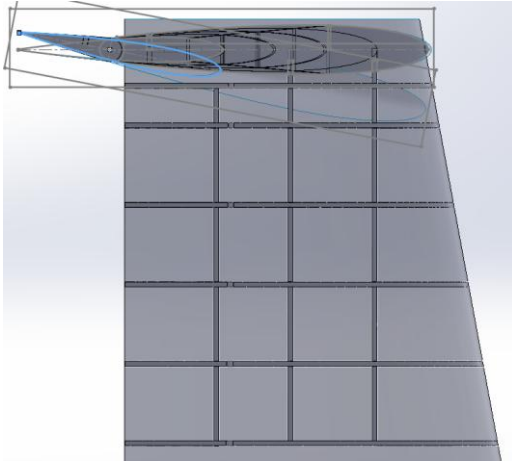


Figure 82 Side view of the final design

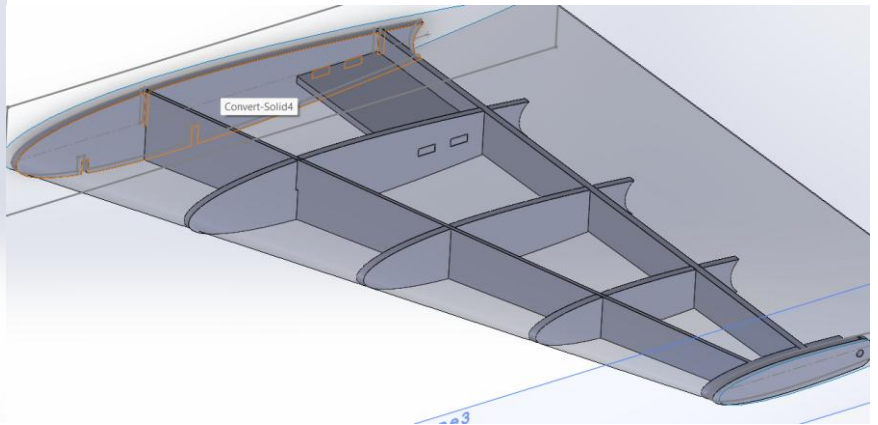


Figure 83 Final Horizontal tail design

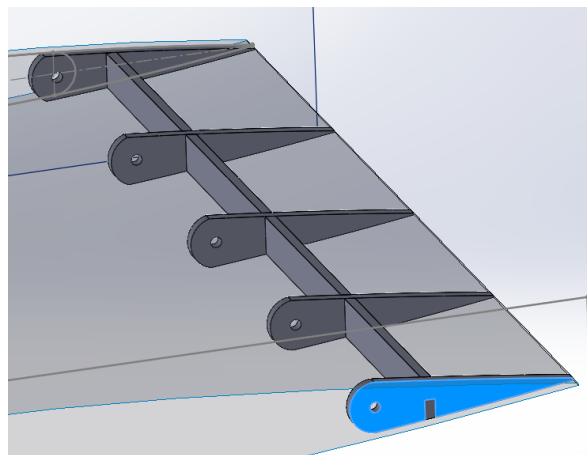


Figure 84 Elevator Final design

An additional spar has been added near the connection to the horizontal tail, helping to retain the aerofoil shape while increasing stiffness and load-carrying capability in this critical region. An extra mounting plate is also installed at the tip of the horizontal tail, allowing the elevator to be attached to the main structure using a pinpoint hinge, which provides smoother and more precise control surface movement. Additional clearance has been incorporated around the hinge line and elevator edges to ensure that the elevator can move freely throughout its full deflection range without binding or interfering with adjacent structure. This configuration not only improves control effectiveness and reliability but also simplifies maintenance and replacement of the control surfaces if needed. Several holes have been added to the bottom plate of vertical tail for connection between the empennage and fuselage.

7.5.7 Build & Manufacturing

The wooden components were manufactured in school using a laser cutting machine. Super glue was applied at the cross-lap joints to ensure they remain secure and structurally sound. A heat-shrink film was then added over the structure to act as the aerodynamic skin and generate lift. After application, the film was heated to tighten and smooth the surface, thereby reducing drag. However, the potential effects of over-shrinking were not fully considered during the design stage. As a result, the leading edge developed a slight wave-like deformation after heating, which can negatively affect aerodynamic performance. In future iterations, the heating process will be more carefully controlled, and additional leading-edge support or pre-curved sheeting may be introduced to maintain a smoother, more stable aerodynamic profile.



Figure 85 Skeleton of the structure before applying wrap

8. Manufacturing Plan (Prepared by Tsang Siu Tung)

8.1 Water Jetting

The carbon fibre plates of mounting the servos are cut by water jet machine. 300mm x 300mm carbon fibre plates with desired thicknesses were bought from Chinese manufacturers. We then cut out the shapes we wanted by using .dxf format files exported from SolidWorks.



Figure 86 Water jetting process with the machine's lid closed

8.2 3D Printing

We would like certain 3D components to be lightweight. As these parts will not be subjected to high impact forces, we have chosen 3D printing (additive manufacturing) as the production method. For prototyping, the material used is PLA+ with an infill of 15%. The machine we used was Bambu Lab X1-Carbon.



Figure 87 3D printing in progress

8.3 Laser Cutting

The plane is mainly made of wood, balsa and plywood plates were manufactured by laser cutting machine. Laser cutting offers high precision and rapid prototyping for manufacturing. It requires little post processing. For 1mm balsa wood plates, we need to use heavy acrylic plates to ensure a flat cutting surface.



Figure 88 Laser cutting in progress